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A Summary of Notations

To facilitate the understanding of the mathematical formulation and optimization process of SFFedMC, we provide a comprehensive summary of the key notations used throughout this paper in Table 1. The symbols are categorized into dimensions, data representations, clustering variables, and causal optimization parameters.

B Theoretical Analysis

B.1 Proof for Theorem 3.1 (Identifiability of Causal Clusters)

Theorem 3.1. *Let $\mathcal{S}$ and $\mathcal{N}$ denote the sets of causal and non-causal features, respectively. Assume the Causal Sufficiency [20] holds. If the global weight vector τ^* minimizes the causal regularizer such that $\sum_v \mathcal{L}_{\text{causal}}^{(v)}(\tau^*) = 0$, then under the weighted distribution P_{τ^*} , the spurious correlation path is blocked. Specifically, the mutual information between non-causal features and the labels vanishes: $I(\mathcal{N}; Y \mid \tau^*) = 0$. Consequently, the optimization of $\mathbf{H}$ relies solely on the causal features $\mathcal{S}$.*

PROOF. We prove this theorem through a multi-stage argument involving structural causal models, RKHS kernel embeddings, and information-theoretic decompositions.

Stage 1: Structural Causal Model and Backdoor Identification.

Recall the SCM in Eq. (2) of the main text. Define the exogenous noise terms as $\epsilon = \{\epsilon_C, \epsilon_S, \epsilon_N, \epsilon_Y\}$ with joint distribution P_ϵ . The observable variables are generated via:

$$\begin{aligned} C &= g_C(\epsilon_C), \quad \mathbf{S}^{(v)} = g_S^{(v)}(\epsilon_S^{(v)}), \\ \mathbf{N}^{(v)} &= g_N^{(v)}(C, \epsilon_N^{(v)}), \quad Y = g_Y(\mathbf{S}^{(1)}, \dots, \mathbf{S}^{(V)}, C, \epsilon_Y), \end{aligned} \quad (1)$$

where $g_\cdot$ are deterministic structural functions. The key causal graph is:

$$Y \xleftarrow{\text{causal}} \mathbf{S}^{(v)} \xrightarrow{\text{generates}} \mathbf{X}^{(v)}, \quad Y \xleftarrow{\text{confounder}} C \xrightarrow{\text{backdoor}} \mathbf{N}^{(v)} \xrightarrow{\text{generates}} \mathbf{X}^{(v)}. \quad (2)$$

Under Pearl’s backdoor criterion [14], the causal effect of $\mathbf{N}$ on Y (i.e., $P(Y \mid \text{do}(\mathbf{N}))$) requires blocking the path $Y \leftarrow C \rightarrow \mathbf{N}$. The adjustment formula is:

$$P(Y \mid \text{do}(\mathbf{N} = \mathbf{n})) = \sum_c P(Y \mid \mathbf{N} = \mathbf{n}, C = c)P(C = c). \quad (3)$$

Table 1: Summary of key notations and their descriptions.

Notation	Description
1. Dimensions and Scalars	
N	Total number of samples (data points)
V	Number of distributed clients (views)
K	Number of target clusters
D_v	Feature dimension of the v -th view
c	Dimension of the latent subspace representation
t	Number of iterations in the optimization process of the algorithm
T	Total number of iterations the algorithm actually runs
$T_{\max}$	Maximum number of iterations allowed
2. Data and Representations	
$\mathbf{X}^{(v)} \in \mathbb{R}^{N \times D_v}$	Raw feature matrix held by the v -th client
$\mathbf{H}^{(v)} \in \mathbb{R}^{N \times c}$	Learned low-rank latent representation of the v -th view
$\mathbf{K}^{(v)} \in \mathbb{R}^{N \times N}$	Local linear kernel matrix, defined as $\mathbf{X}^{(v)} (\mathbf{X}^{(v)})^\top$
$\mathbf{M}^{(v)}$	Auxiliary matrix for constructing the Procrustes problem
3. Clustering and Consensus	
$\mathbf{C}^{(v)} \in \mathbb{R}^{K \times c}$	Local cluster centroid matrix of the v -th view
$\mathbf{y}^{(v)} \in \{1, \dots, K\}^N$	Local pseudo-label vector of the v -th view
$\mathbf{Y}^{(v)} \in \{0, 1\}^{N \times K}$	One-hot encoding matrix of $\mathbf{y}^{(v)}$
$\mathbf{C}_c \in \mathbb{R}^{K \times c}$	Global consensus centroid matrix on the server
$\mathbf{y}_c \in \{1, \dots, K\}^N$	Global consensus cluster label vector
$\mathbf{Y}_c \in \{0, 1\}^{N \times K}$	One-hot encoding matrix of $\mathbf{y}_c$
C_k	Set of indices belonging to the k -th cluster
4. Causal Inference and Optimization	
$\boldsymbol{\tau} \in \mathbb{R}^N$	Global sample weight vector (Causal De-confounding Weights)
$\Gamma(\boldsymbol{\tau}) \in \mathbb{R}^{N \times N}$	Diagonal weight matrix, $\Gamma(\boldsymbol{\tau}) = \text{diag}(\boldsymbol{\tau})$
C_v	Treatment group defined on view v for MMD
$\mathcal{T}_v$	Control group defined on view v for MMD
$\mathcal{L}_{\text{causal}}$	Causal regularization loss (Multi-View MMD)
λ	Weight for local representation approximation term
β	Weight for global consensus integration term
γ	Weight for causal regularization term
η	Learning rate for the projected gradient descent on $\boldsymbol{\tau}$

Expanding the right-hand side using the chain rule:

$$P(Y \mid do(\mathbf{N})) = \sum_c P(Y \mid \mathbf{N}, C = c) P(C = c) \quad (4)$$

$$= \sum_c \frac{P(Y, \mathbf{N}, C = c)}{P(\mathbf{N}, C = c)} P(C = c) \quad (5)$$

$$= \sum_c \frac{P(Y, \mathbf{N}, C = c)}{P(\mathbf{N} \mid C = c)}. \quad (6)$$

However, in Eq. (6), if $P(N | C = c)$ is constant w.r.t. c (i.e., $N \perp C$), then:

$$P(Y | do(N)) = P(Y), \quad (7)$$

establishing that N has no causal effect on Y under the intervention. This motivates our re-weighting strategy.

Stage 2: Re-weighted Distribution and Kernel Mean Embedding.

Since C is latent, we cannot directly condition on it. Instead, we construct a re-weighted empirical distribution P_{τ} via importance weights $\tau \in \Delta^{N-1}$:

$$P_{\tau}(\mathcal{E}) = \sum_{i=1}^N \tau_i \mathbb{I}[z_i \in \mathcal{E}], \quad \text{for any event } \mathcal{E}, \quad (8)$$

where $z_i = (\mathbf{h}_i, y_i)$ is the latent representation-label pair for sample i . Our goal is to find τ^* such that under P_{τ^*} , the confounder C is balanced across treatment groups defined by N .

Define the treatment assignment $A : \mathcal{X} \rightarrow \{0, 1\}$ via Eq. (5) in the main text. The covariate balance condition is:

$$P_{\tau^*}(\mathbf{H} | A = 1) = P_{\tau^*}(\mathbf{H} | A = 0) = P_{\tau^*}(\mathbf{H}). \quad (9)$$

In the RKHS $\mathcal{H}$ with kernel $k(\cdot, \cdot)$, this translates to matching kernel mean embeddings:

$$\mu_{\mathcal{T}}(\tau^*) := \mathbb{E}_{P_{\tau^*}}[\phi(\mathbf{H}) | A = 1] \stackrel{!}{=} \mathbb{E}_{P_{\tau^*}}[\phi(\mathbf{H}) | A = 0] =: \mu_C(\tau^*). \quad (10)$$

By the reproducing property of RKHS, the squared MMD is:

$$\begin{aligned} \text{MMD}^2(P_{\mathcal{T}}, P_C) &= \|\mu_{\mathcal{T}} - \mu_C\|_{\mathcal{H}}^2 \\ &= \mathbb{E}_{i,j \in \mathcal{T}}[k(\mathbf{h}_i, \mathbf{h}_j)] + \mathbb{E}_{i,j \in C}[k(\mathbf{h}_i, \mathbf{h}_j)] - 2\mathbb{E}_{i \in \mathcal{T}, j \in C}[k(\mathbf{h}_i, \mathbf{h}_j)]. \end{aligned} \quad (11)$$

For a characteristic kernel (e.g., Gaussian or linear), $\text{MMD}^2 = 0$ if and only if $P_{\mathcal{T}} = P_C$ [10]. Thus, if $\mathcal{L}_{\text{causal}}^{(v)}(\tau^*) = 0$ for all v , Eq. (9) is satisfied.

Stage 3: Conditional Independence via Balancing Weights.

Under the balanced distribution P_{τ^*} , we now show $N \perp Y | \tau^*$. First, decompose the joint distribution:

$$P_{\tau^*}(Y, N) = \sum_c P_{\tau^*}(Y, N, C = c) = \sum_c P_{\tau^*}(Y | C = c) P_{\tau^*}(N | C = c) P_{\tau^*}(C = c). \quad (12)$$

By the SCM in Eq. (1), N is a deterministic function of C (up to noise ϵ_N), and Y depends on C and S (up to noise ϵ_Y). However, $S \perp C$ by construction (causal sufficiency assumption [14]). Thus:

$$P(Y, N | C) = P(Y | S, C) P(N | C) P(S). \quad (13)$$

Now, the key step: under the re-weighted distribution P_{τ^*} , if the treatment groups $\mathcal{T}, C$ (partitioned by N via Eq. (5) in the main text) have balanced confounder distributions, then:

$$P_{\tau^*}(C | A = 1) = P_{\tau^*}(C | A = 0) = P_{\tau^*}(C). \quad (14)$$

Since A is a function of N (specifically, $A = \mathbb{I}[N \geq \text{median}(N)]$), this implies:

$$P_{\tau^*}(C | N) = P_{\tau^*}(C). \quad (15)$$

Substituting Eq. (15) into Eq. (12):

$$P_{\tau^*}(Y, N) = \sum_c P_{\tau^*}(Y | C = c) P_{\tau^*}(N | C = c) P_{\tau^*}(C = c) \quad (16)$$

$$= \sum_c P_{\tau^*}(Y | C = c) P_{\tau^*}(C = c) \cdot P_{\tau^*}(N) \quad (\text{by Eq. (15)}) \quad (17)$$

$$= P_{\tau^*}(Y) \cdot P_{\tau^*}(N). \quad (18)$$

Thus, $Y \perp N | P_{\tau^*}$.

Stage 4: Information-Theoretic Derivation.

To rigorously establish $I(N; Y | \tau^*) = 0$, we employ the mutual information decomposition:

$$\begin{aligned} I(N; Y | \tau^*) &= \sum_{y \in \mathcal{Y}} \sum_{\mathbf{n} \in \mathcal{N}} P_{\tau^*}(y, \mathbf{n}) \log \frac{P_{\tau^*}(y, \mathbf{n})}{P_{\tau^*}(y) P_{\tau^*}(\mathbf{n})} \\ &= \mathbb{E}_{P_{\tau^*}(Y, N)} \left[\log \frac{P_{\tau^*}(Y, N)}{P_{\tau^*}(Y) P_{\tau^*}(N)} \right]. \end{aligned} \quad (19)$$

From Eq. (18), we have $P_{\tau^*}(Y, \mathbf{N}) = P_{\tau^*}(Y)P_{\tau^*}(\mathbf{N})$. Substituting:

$$I(\mathcal{N}; Y | \tau^*) = \mathbb{E}_{P_{\tau^*}(Y, \mathbf{N})} \left[\log \frac{P_{\tau^*}(Y)P_{\tau^*}(\mathbf{N})}{P_{\tau^*}(Y)P_{\tau^*}(\mathbf{N})} \right] \quad (20)$$

$$= \mathbb{E}_{P_{\tau^*}(Y, \mathbf{N})} [\log 1] = 0. \quad (21)$$

Stage 5: Generalization to Multi-View Setting.

The above argument extends to multiple views by noting that $\mathcal{L}_{causal}(\tau^*) = \sum_{v=1}^V \mathcal{L}_{causal}^{(v)}(\tau^*) = 0$ implies $\mathcal{L}_{causal}^{(v)}(\tau^*) = 0$ for each v (since each term is non-negative). Therefore, the independence $\mathbf{N}^{(v)} \perp Y | P_{\tau^*}$ holds for all views simultaneously. The overall mutual information is:

$$I(\{\mathcal{N}^{(v)}\}_{v=1}^V; Y | \tau^*) = \sum_{v=1}^V I(\mathcal{N}^{(v)}; Y | \tau^*, \mathcal{N}^{(-v)}) = 0, \quad (22)$$

where $\mathcal{N}^{(-v)}$ denotes non-causal features from views other than v . This completes the proof. $\square$

B.2 Proof for Theorem 3.2 (Convergence)

Theorem 3.2. *The objective function $\mathcal{J}(\Theta)$ in Eq. (8) of the main text is upper-bounded. The alternating optimization algorithm of SFFedMC generates a sequence of parameter states $\{\Theta^{(t)}\}_{t=1}^\infty$ such that the objective value is monotonically non-decreasing (i.e., $\mathcal{J}(\Theta^{(t+1)}) \geq \mathcal{J}(\Theta^{(t)})$). The sequence converges to a local stationary point Θ^* .*

PROOF. Let $\mathcal{J}(\Theta)$ be the global objective function defined in Eq. (8) of the main text. We establish convergence through a multi-part analysis of the Block Coordinate Ascent (BCA) scheme, demonstrating: (i) monotonicity of each sub-block update, (ii) boundedness of the objective sequence, and (iii) convergence to a stationary point via Zangwill's Global Convergence Theorem [16].

Part 1: Client-Side Representation Update (Orthogonal Procrustes Problem).

Fix view v and all parameters except $\mathbf{H}^{(v)}$. The sub-problem is:

$$\mathcal{P}_H : \max_{\mathbf{H}^{(v)} \in \mathcal{S}\sqcup(N, c)} f_H(\mathbf{H}^{(v)}) := \text{Tr}[(\mathbf{H}^{(v)})^\top \Gamma(\tau) \mathbf{K}^{(v)} \mathbf{H}^{(v)}] + \lambda \text{Tr}[(\mathbf{H}^{(v)})^\top \mathbf{Y}^{(v)} \mathbf{C}^{(v)}], \quad (23)$$

where $\mathcal{S}\sqcup(N, c) = \{\mathbf{H} \in \mathbb{R}^{N \times c} : \mathbf{H}^\top \mathbf{H} = \mathbf{I}_c\}$ is the Stiefel manifold. Define auxiliary matrix $\mathbf{M}^{(v)} = \Gamma(\tau) \mathbf{K}^{(v)} \mathbf{H}^{(v)} + \lambda \mathbf{Y}^{(v)} \mathbf{C}^{(v)}$. Then:

$$f_H(\mathbf{H}^{(v)}) = \text{Tr}[(\mathbf{H}^{(v)})^\top \mathbf{M}^{(v)}] + \text{const}. \quad (24)$$

Lemma 3.4 (Von Neumann's Trace Inequality). For any matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1(\mathbf{A}) \geq \dots \geq \sigma_r(\mathbf{A})$ and $\sigma_1(\mathbf{B}) \geq \dots \geq \sigma_r(\mathbf{B})$:

$$\text{Tr}(\mathbf{A}^\top \mathbf{B}) \leq \sum_{i=1}^r \sigma_i(\mathbf{A}) \sigma_i(\mathbf{B}), \quad (25)$$

with equality if and only if $\mathbf{A} = \mathbf{U} \Sigma_A \mathbf{V}^\top$ and $\mathbf{B} = \mathbf{U} \Sigma_B \mathbf{V}^\top$ share the same left and right singular subspaces.

Apply Lemma 3.4 to $\mathcal{P}_H$: let $\mathbf{M}^{(v)} = \mathbf{U}_M \Sigma_M \mathbf{V}_M^\top$ be the SVD. Then:

$$\text{Tr}[(\mathbf{H}^{(v)})^\top \mathbf{M}^{(v)}] \leq \sum_{i=1}^c \sigma_i(\mathbf{H}^{(v)}) \sigma_i(\mathbf{M}^{(v)}). \quad (26)$$

Since $\mathbf{H}^{(v)} \in \mathcal{S}\sqcup(N, c)$, all singular values of $\mathbf{H}^{(v)}$ are 1. Thus:

$$\max_{\mathbf{H} \in \mathcal{S}\sqcup(N, c)} \text{Tr}[(\mathbf{H})^\top \mathbf{M}^{(v)}] = \sum_{i=1}^c \sigma_i(\mathbf{M}^{(v)}), \quad (27)$$

achieved at $\mathbf{H}_{\text{new}}^{(v)} = \mathbf{U}_M \mathbf{V}_M^\top$ (the orthogonal Procrustes solution). Therefore:

$$f_H(\mathbf{H}_{\text{new}}^{(v)}) = \sum_{i=1}^c \sigma_i(\mathbf{M}^{(v)}) \geq f_H(\mathbf{H}_{\text{old}}^{(v)}), \quad (28)$$

which implies $\mathcal{J}(\Theta^{(t+1)}) \geq \mathcal{J}(\Theta^{(t)})$ after the $\mathbf{H}$ -update.

Part 2: Client-Side Label Update (Discrete Optimization).

Fixing $\mathbf{H}^{(v)}$, the local pseudo-label update is:

$$\mathbf{y}_i^{(v)} = \arg \max_{k \in [K]} \left\{ \lambda \langle \mathbf{H}_{i,:}^{(v)}, \mathbf{C}_{k,:}^{(v)} \rangle + \beta \langle [\mathbf{C}_c]_{y_c(i),:}, \mathbf{C}_{k,:}^{(v)} \rangle \right\}. \quad (29)$$

Since this is a greedy argmax, it trivially satisfies:

$$\sum_{i=1}^N \left[\lambda \langle \mathbf{H}_{i,:}^{(v)}, \mathbf{C}_{\mathbf{y}_i^{(v)},:}^{(v)} \rangle + \beta \langle [\mathbf{C}_c]_{y_c(i),:}, \mathbf{C}_{\mathbf{y}_i^{(v)},:}^{(v)} \rangle \right]_{\text{new}} \geq [\dots]_{\text{old}}. \quad (30)$$

Thus, the objective $\mathcal{J}$ is non-decreasing after label updates.

Part 3: Server-Side Consensus Centroid Update (Orthogonal Procrustes on Transpose).

The server optimizes $\mathbf{C}_c$ given fixed $\mathbf{H}_c = [\mathbf{Y}^{(1)}\mathbf{C}^{(1)}, \dots, \mathbf{Y}^{(V)}\mathbf{C}^{(V)}]$ and $\mathbf{Y}_c$:

$$\max_{\mathbf{C}_c \mathbf{C}_c^\top = \mathbf{I}_K} \text{Tr}[(\mathbf{C}_c)^\top (\mathbf{Y}_c)^\top \mathbf{H}_c]. \quad (31)$$

Let $\mathbf{M}_c = \mathbf{H}_c^\top \mathbf{Y}_c \in \mathbb{R}^{(\sum c_v) \times K}$, and rewrite:

$$\max_{\mathbf{C}_c} \text{Tr}[\mathbf{C}_c^\top \mathbf{M}_c^\top] = \max_{\mathbf{C}_c} \text{Tr}[(\mathbf{M}_c \mathbf{C}_c)^\top] = \max_{\mathbf{C}_c} \text{Tr}[\mathbf{M}_c \mathbf{C}_c]. \quad (32)$$

By the transpose duality of the Procrustes problem, if $\mathbf{M}_c = \mathbf{U}_c \Sigma_c \mathbf{V}_c^\top$, then:

$$\mathbf{C}_c = \mathbf{V}_c \mathbf{U}_c^\top \Rightarrow \text{Tr}[\mathbf{M}_c \mathbf{C}_c] = \text{Tr}[\Sigma_c] = \sum_{i=1}^{\min(K, \sum c_v)} \sigma_i(\mathbf{M}_c), \quad (33)$$

which is the global maximum. Thus, $\mathcal{J}(\Theta^{(t+1)}) \geq \mathcal{J}(\Theta^{(t)})$ after the $\mathbf{C}_c$ -update.

Part 4: Server-Side Consensus Label Update (Weighted Greedy Assignment).

Given $\mathbf{C}_c$ and $\boldsymbol{\tau}$, the server updates:

$$(\mathbf{y}_c)_i = \arg \max_{k \in [K]} \{\tau_i \cdot [\mathbf{H}_c \mathbf{C}_c^\top]_{ik}\}. \quad (34)$$

Since this is also a greedy argmax over discrete choices, the objective term $\beta \text{Tr}[\mathbf{H}_c^\top \mathbf{Y}_c \mathbf{C}_c]$ satisfies:

$$\beta \sum_{i=1}^N \tau_i [\mathbf{H}_c \mathbf{C}_c^\top]_{i, y_c(i)}^{\text{new}} \geq \beta \sum_{i=1}^N \tau_i [\mathbf{H}_c \mathbf{C}_c^\top]_{i, y_c(i)}^{\text{old}}, \quad (35)$$

ensuring monotonic increase in $\mathcal{J}$.

Part 5: Server-Side Causal Weight Update (Constrained Convex Optimization).

The weight optimization is:

$$\min_{\boldsymbol{\tau} \in \Delta^{N-1}} \mathcal{L}_{\text{causal}}(\boldsymbol{\tau}) = \sum_{v=1}^V \|\boldsymbol{\mu}_{\mathcal{T}_v}(\boldsymbol{\tau}) - \boldsymbol{\mu}_{C_v}(\boldsymbol{\tau})\|_{\mathcal{H}}^2. \quad (36)$$

Expand the squared norm using the bilinearity of the inner product:

$$\begin{aligned} \mathcal{L}_{\text{causal}}^{(v)}(\boldsymbol{\tau}) &= \langle \boldsymbol{\mu}_{\mathcal{T}}, \boldsymbol{\mu}_{\mathcal{T}} \rangle - 2\langle \boldsymbol{\mu}_{\mathcal{T}}, \boldsymbol{\mu}_C \rangle + \langle \boldsymbol{\mu}_C, \boldsymbol{\mu}_C \rangle \\ &= \frac{\boldsymbol{\tau}^\top (\mathbf{a} \odot \mathbf{K} \odot \mathbf{a}) \boldsymbol{\tau}}{(\mathbf{1}^\top (\boldsymbol{\tau} \odot \mathbf{a}))^2} - 2 \frac{\boldsymbol{\tau}^\top (\mathbf{a} \odot \mathbf{K} \odot (\mathbf{1} - \mathbf{a})) \boldsymbol{\tau}}{(\mathbf{1}^\top (\boldsymbol{\tau} \odot \mathbf{a}))(\mathbf{1}^\top (\boldsymbol{\tau} \odot (\mathbf{1} - \mathbf{a})))} \\ &\quad + \frac{\boldsymbol{\tau}^\top ((\mathbf{1} - \mathbf{a}) \odot \mathbf{K} \odot (\mathbf{1} - \mathbf{a})) \boldsymbol{\tau}}{(\mathbf{1}^\top (\boldsymbol{\tau} \odot (\mathbf{1} - \mathbf{a})))^2}, \end{aligned} \quad (37)$$

where $\mathbf{K} = \mathbf{H}^{(v)}(\mathbf{H}^{(v)})^\top$ is the Gram matrix, and $\odot$ denotes Hadamard product. Each term in Eq. (37) is a ratio of quadratic forms, which is quasi-convex in $\boldsymbol{\tau}$ (see Lemma 3.5 below).

Lemma 3.5 (Quasi-Convexity of Ratio of Quadratic Forms). Let $f(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{(\mathbf{b}^\top \mathbf{x})^2}$ where $\mathbf{A} \succeq \mathbf{0}$ (positive semidefinite) and $\mathbf{b}^\top \mathbf{x} > 0$ for all $\mathbf{x}$ in the feasible region. Then f is quasi-convex on $\{\mathbf{x} : \mathbf{b}^\top \mathbf{x} > 0\}$.

Proof sketch of Lemma 3.5: For any $\alpha \in \mathbb{R}$, the sublevel set $\{\mathbf{x} : f(\mathbf{x}) \leq \alpha\}$ is equivalent to $\{\mathbf{x} : \mathbf{x}^\top (\mathbf{A} - \alpha \mathbf{b} \mathbf{b}^\top) \mathbf{x} \leq 0\}$, which is a convex set (intersection of halfspaces). Thus, f is quasi-convex. $\square$

By Lemma 3.5 and the fact that sums of quasi-convex functions remain quasi-convex, $\mathcal{L}_{\text{causal}}(\boldsymbol{\tau})$ is quasi-convex on the simplex Δ^{N-1} . Projected Gradient Descent (PGD) on quasi-convex functions over convex sets is guaranteed to converge to the global minimum [2]. Thus, each PGD step satisfies:

$$\mathcal{L}_{\text{causal}}(\boldsymbol{\tau}^{(t+1)}) \leq \mathcal{L}_{\text{causal}}(\boldsymbol{\tau}^{(t)}), \quad (38)$$

which translates to $\mathcal{J}(\Theta^{(t+1)}) \geq \mathcal{J}(\Theta^{(t)})$ after the $\boldsymbol{\tau}$ -update (since $\mathcal{L}_{\text{causal}}$ enters $\mathcal{J}$ with a negative sign).

Part 6: Boundedness of the Objective Function.

We now show that $\mathcal{J}(\Theta)$ is bounded from above. The objective consists of three types of terms:

(1) **Kernel alignment terms:**

$$\text{Tr}[(\mathbf{H}^{(v)})^\top \boldsymbol{\Gamma}(\boldsymbol{\tau}) \mathbf{K}^{(v)} \mathbf{H}^{(v)}] \leq \|\boldsymbol{\Gamma}(\boldsymbol{\tau})\|_{\text{op}} \|\mathbf{K}^{(v)}\|_{\text{op}} \|\mathbf{H}^{(v)}\|_F^2 = \|\boldsymbol{\Gamma}(\boldsymbol{\tau})\|_{\text{op}} \|\mathbf{K}^{(v)}\|_{\text{op}} \cdot c, \quad (39)$$

where $\|\boldsymbol{\Gamma}(\boldsymbol{\tau})\|_{\text{op}} = \max_i \tau_i \leq 1$ (since $\boldsymbol{\tau} \in \Delta^{N-1}$), and $\|\mathbf{K}^{(v)}\|_{\text{op}} = \|\mathbf{X}^{(v)}\|_{\text{op}}^2 < \infty$ (bounded by data). Thus, this term is $O(c)$.

(2) **Trace approximation terms:**

$$\text{Tr}[(\mathbf{H}^{(v)})^\top \mathbf{Y}^{(v)} \mathbf{C}^{(v)}] \leq \|\mathbf{H}^{(v)}\|_F \|\mathbf{Y}^{(v)}\|_F \|\mathbf{C}^{(v)}\|_F = \sqrt{N_c} \cdot \sqrt{N} \cdot \sqrt{K_c} = N_c \sqrt{K_c}. \quad (40)$$

(3) **MMD regularization:** Since $\mathcal{L}_{\text{causal}}$ is a squared RKHS norm, it is non-negative and bounded by the kernel radius: $\mathcal{L}_{\text{causal}} \leq 4 \sup_{\mathbf{h}} \|\phi(\mathbf{h})\|_{\mathcal{H}}^2 < \infty$.

Combining all terms: $\mathcal{J}(\Theta) \leq C_1 Vc + C_2 V N c \sqrt{K} + C_3 < \infty$ for constants C_1, C_2, C_3 depending on data norms and hyperparameters.

Part 7: Convergence to Stationary Point.

By Zangwill's Global Convergence Theorem [16], a block coordinate ascent algorithm converges to a stationary point if:

- (1) The objective is continuous in each block (all blocks involve smooth or discrete updates).
- (2) The objective is non-decreasing (shown in **Parts 1–5**).
- (3) The objective is bounded (shown in **Part 6**).
- (4) Each subproblem admits a unique or closed-form solution (all use SVD or argmax).

Therefore, the sequence $\{\mathcal{J}(\Theta^{(t)})\}_{t=1}^{\infty}$ converges to a limit $\mathcal{J}^* = \lim_{t \rightarrow \infty} \mathcal{J}(\Theta^{(t)})$. By the Monotone Convergence Theorem, since the sequence is bounded and monotonically non-decreasing, it must converge. Furthermore, since all subproblems are solved exactly, the limit point Θ^* satisfies the first-order optimality conditions (KKT conditions for constrained subproblems), making it a local stationary point of $\mathcal{J}$.

This completes the proof. $\square$

B.3 Proof for Lemma 3.3 (Privacy Preservation)

Lemma 3.3. *During the execution of SFFedMC, the raw data matrices $\mathbf{X}^{(v)}$ and the local kernel matrices $\mathbf{K}^{(v)}$ are strictly confined to the local clients. The information exchanged between clients and the server is limited to the low-rank latent representations $\mathbf{H}^{(v)}$, centroids $\mathbf{C}^{(v)}$, and scalar weights τ , preventing the reconstruction of raw features.*

PROOF. We establish privacy through a multi-layered information-theoretic analysis of the communication protocol, demonstrating that the transmitted messages constitute an under-determined system with exponentially many solutions, rendering exact reconstruction computationally infeasible.

Stage 1: Forward Pass Communication Analysis (Client $\rightarrow$ Server).

Each client v transmits two pieces of information:

$$\mathcal{I}_{\text{up}}^{(v)} = \{\mathbf{y}^{(v)} \in \mathbb{Z}_K^N, \mathbf{C}^{(v)} \in \mathbb{R}^{K \times c}\}, \quad (41)$$

where $\mathbf{y}^{(v)}$ is the integer label vector (requires $N \log_2 K$ bits), and $\mathbf{C}^{(v)}$ is the local centroid matrix (requires Kc floats). The total uplink communication is $O(N \log K + Kc)$ scalars.

Question: Can an adversary on the server reconstruct $\mathbf{X}^{(v)} \in \mathbb{R}^{N \times D_v}$ from $\mathcal{I}_{\text{up}}^{(v)}$?

Reconstruction Model. Suppose the data generation follows $\mathbf{X}^{(v)} = \mathbf{H}^{(v)} \mathbf{P}^{(v)}$, where $\mathbf{H}^{(v)} \in \mathbb{R}^{N \times c}$ is the latent representation and $\mathbf{P}^{(v)} \in \mathbb{R}^{c \times D_v}$ is the projection basis. The server knows that:

$$\mathbf{Y}^{(v)} \mathbf{C}^{(v)} \approx \mathbf{H}^{(v)}, \quad (42)$$

where $\mathbf{Y}^{(v)} \in \{0, 1\}^{N \times K}$ is the one-hot encoding of $\mathbf{y}^{(v)}$. However, this only provides:

$$\mathbf{X}^{(v)} \approx \mathbf{Y}^{(v)} \mathbf{C}^{(v)} \mathbf{P}^{(v)}. \quad (43)$$

Under-Determinedness Analysis. The reconstruction problem requires solving for $(\mathbf{H}^{(v)}, \mathbf{P}^{(v)})$ such that $\mathbf{X}^{(v)} = \mathbf{H}^{(v)} \mathbf{P}^{(v)}$ given only $\mathbf{Y}^{(v)} \mathbf{C}^{(v)}$. Count the degrees of freedom:

- **Unknowns:** $\mathbf{H}^{(v)}$ has Nc free parameters; $\mathbf{P}^{(v)}$ has cD_v parameters. Total: $Nc + cD_v = c(N + D_v)$ unknowns.
- **Constraints:** From $\mathbf{Y}^{(v)} \mathbf{C}^{(v)} = \mathbf{H}^{(v)}$, we get Nc equations. However, $\mathbf{Y}^{(v)}$ has rank at most K , so the effective constraints are only Kc (since $\mathbf{Y}^{(v)} \mathbf{C}^{(v)}$ lies in a K -dimensional subspace).
- **Deficit:** The system has $c(N + D_v)$ unknowns but only $\min(Nc, Kc)$ effective constraints. For typical settings where $D_v \gg K$ (high-dimensional features), the deficit is:

$$c(N + D_v) - Kc = c(N + D_v - K) \gg 0. \quad (44)$$

Lemma 3.6 (Indeterminacy of Factorization). For any given $\mathbf{H}_0^{(v)}$ satisfying $\mathbf{Y}^{(v)} \mathbf{C}^{(v)} = \mathbf{H}_0^{(v)}$, there exist infinitely many pairs $(\mathbf{H}^{(v)}, \mathbf{P}^{(v)})$ that produce the same $\mathbf{X}^{(v)}$. Specifically, for any invertible matrix $\mathbf{Q} \in \mathbb{R}^{c \times c}$:

$$\mathbf{X}^{(v)} = \mathbf{H}_0^{(v)} \mathbf{P}_0^{(v)} = (\mathbf{H}_0^{(v)} \mathbf{Q})(\mathbf{Q}^{-1} \mathbf{P}_0^{(v)}). \quad (45)$$

The set of all such factorizations forms a manifold of dimension $c^2 - c$ (the dimension of $\text{GL}(c)$ modulo scaling).

Proof of Lemma 3.6: This follows directly from the non-identifiability of matrix factorization. The transformation $(\mathbf{H}, \mathbf{P}) \mapsto (\mathbf{H}\mathbf{Q}, \mathbf{Q}^{-1}\mathbf{P})$ leaves the product invariant, and the general linear group $\text{GL}(c)$ has dimension c^2 . Modulo the scalar invariance $\mathbf{Q} = \alpha \mathbf{I}$ (dimension 1), the quotient space has dimension $c^2 - 1 \approx c^2$ for large c . $\square$

Entropy-Based Argument. Define the *reconstruction entropy*:

$$H(\mathbf{X}^{(v)} | \mathcal{I}_{\text{up}}^{(v)}) = H(\mathbf{X}^{(v)}) - I(\mathbf{X}^{(v)}; \mathcal{I}_{\text{up}}^{(v)}), \quad (46)$$

where $I(\cdot; \cdot)$ is mutual information. Since $\mathcal{I}_{\text{up}}^{(v)}$ contains $O(N \log K + Kc)$ bits, and $\mathbf{X}^{(v)}$ contains ND_v floats (approximately $32ND_v$ bits):

$$\begin{aligned} H(\mathbf{X}^{(v)} | \mathcal{I}_{\text{up}}^{(v)}) &\geq H(\mathbf{X}^{(v)}) - |\mathcal{I}_{\text{up}}^{(v)}| \\ &\geq 32ND_v - (N \log_2 K + 32Kc) \\ &\geq N(32D_v - \log_2 K) - 32Kc. \end{aligned} \quad (47)$$

For $D_v \gg K$ (e.g., $D_v = 10^3, K = 10$), the conditional entropy is still exponentially large in N , meaning the adversary's uncertainty about $\mathbf{X}^{(v)}$ remains nearly maximal.

Stage 2: Backward Pass Communication (Server $\rightarrow$ Client).

The server broadcasts:

$$\mathcal{I}_{\text{down}} = \{\mathbf{y}_c \in \mathbb{Z}_K^N, \mathbf{C}_c \in \mathbb{R}^{K \times (\sum c_v)}, \boldsymbol{\tau} \in \mathbb{R}_+^N\}. \quad (48)$$

These are all aggregated statistics that do not directly reveal any single client's raw features:

- (1) **Consensus labels $\mathbf{y}_c$** : Derived from weighted voting across all clients (Eq. (16) in the main text). Each label depends on contributions from all V views, making it impossible to isolate view v 's contribution without knowing the others.
- (2) **Global centroids $\mathbf{C}_c$** : Computed via SVD of $\mathbf{H}_c = [\mathbf{Y}^{(1)}\mathbf{C}^{(1)}, \dots, \mathbf{Y}^{(V)}\mathbf{C}^{(V)}]$ (Eq. (15) in the main text). Since $\mathbf{C}_c$ is derived from a low-rank approximation (rank K) of a concatenated matrix, it loses information about individual views through the dimensionality reduction.
- (3) **Causal weights $\boldsymbol{\tau}$** : These are scalar reweighting factors optimized via MMD minimization (Eq. (17) in the main text). They encode distributional information across views but do not encode feature values.

Differential Privacy Perspective (Informal). Although we do not claim formal differential privacy, we can bound the information leakage. Define the *per-sample influence* of client v on the global centroid:

$$\Delta_i^{(v)}(\mathbf{C}_c) = \|\mathbf{C}_c(\mathcal{D}) - \mathbf{C}_c(\mathcal{D} \setminus \{i\})\|_F, \quad (49)$$

where $\mathcal{D}$ is the dataset. By the stability of SVD [17], if $\mathbf{C}_c$ is the top- K subspace of an $N \times (\sum c_v)$ matrix, then:

$$\Delta_i^{(v)}(\mathbf{C}_c) \leq \frac{\|\mathbf{Y}_i^{(v)}\mathbf{C}^{(v)}\|_2}{\sigma_K(\mathbf{H}_c) \cdot N} \leq \frac{\sqrt{c}}{N\sigma_K}, \quad (50)$$

where σ_K is the K -th singular value. For large N , the per-sample influence vanishes as $O(1/N)$, implying that removing any single sample has a negligible effect on the global model.

Stage 3: Gradient-Based Attack Resistance.

Modern gradient inversion attacks [21] attempt to reconstruct $\mathbf{X}^{(v)}$ by solving:

$$\min_{\mathbf{X}^{(v)}} \left\| \nabla_{\mathbf{X}} \mathcal{L}(\mathbf{X}^{(v)}) - \mathbf{g}_{\text{received}} \right\|^2, \quad (51)$$

where $\mathbf{g}_{\text{received}}$ is the gradient transmitted from the client. However, in SFFedMC:

- **No gradients are transmitted.** Clients send only discrete labels $\mathbf{y}^{(v)}$ and low-rank centroids $\mathbf{C}^{(v)}$, not gradient vectors.
- **Latent representation is lossy.** The mapping $\mathbf{X}^{(v)} \rightarrow \mathbf{H}^{(v)}$ via SVD discards the singular values $\Sigma^{(v)}$ and right singular vectors $\mathbf{V}^{(v)}$. Even if an adversary obtains $\mathbf{H}^{(v)} = \mathbf{U}^{(v)}$, reconstructing $\mathbf{X}^{(v)} = \mathbf{U}^{(v)}\Sigma^{(v)}(\mathbf{V}^{(v)})^\top$ requires guessing $\Sigma^{(v)}$ (order c unknowns) and $\mathbf{V}^{(v)}$ (order cD_v unknowns), which is exponentially hard.

Formal Statement of Reconstruction Hardness. Define the *reconstruction problem*:

Problem B.1 (Data Reconstruction). *Given $(\mathbf{y}^{(v)}, \mathbf{C}^{(v)}) \in \mathbb{Z}_K^N \times \mathbb{R}^{K \times c}$, recover $\mathbf{X}^{(v)} \in \mathbb{R}^{N \times D_v}$ such that:*

$$\mathbf{X}^{(v)} \approx \mathbf{Y}^{(v)}\mathbf{C}^{(v)}\mathbf{P}^{(v)}. \quad (52)$$

Theorem (Computational Hardness). Assuming $D_v \geq c^2$ and the projection matrix $\mathbf{P}^{(v)}$ is uniformly distributed over the Stiefel manifold $S\sqcup(c, D_v)$, the problem of recovering $\mathbf{X}^{(v)}$ from $(\mathbf{y}^{(v)}, \mathbf{C}^{(v)})$ is computationally equivalent to solving an under-determined linear system with $\Omega(D_v - K)$ free parameters, which has no unique solution.

Proof: The number of unknown parameters in $\mathbf{P}^{(v)}$ is $cD_v - \frac{c(c+1)}{2}$ (accounting for orthonormality constraints). The number of constraints from $\mathbf{Y}^{(v)}\mathbf{C}^{(v)} = \mathbf{H}^{(v)}$ is at most Kc . For $D_v \gg K$, the system is severely under-determined, with $cD_v - Kc \gg 0$ free dimensions. By information theory, the space of compatible solutions has measure proportional to $\exp(c(D_v - K))$, making brute-force search infeasible. $\square$

Conclusion. Combining *Stages 1–3*, we conclude that: (i) raw data $\mathbf{X}^{(v)}$ is never transmitted; (ii) the transmitted information forms an under-determined system with exponentially many solutions; (iii) gradient-based attacks are inapplicable; and (iv) the per-sample influence on global parameters vanishes as $O(1/N)$. Therefore, SFFedMC preserves privacy in the honest-but-curious adversary model. $\square$

C Detailed Derivation of MMD Gradient

In this section, we provide the complete derivation of the gradient $\nabla_{\tau} \mathcal{L}_{causal}^{(v)}(\tau)$ used in the causal weight update (Eq. (22) in the main text). This derivation is technically involved due to the ratio terms arising from weighted mean embeddings.

Setup. For $i \in \mathcal{T}$, define auxiliary notation: $\mathbf{w}_{\mathcal{T}} := \tau \odot \mathbf{a}$, $\mathbf{w}_{\mathcal{C}} := \tau \odot (\mathbf{1} - \mathbf{a})$, $Z_{\mathcal{T}} := \sum_{i \in \mathcal{T}} \tau_i$, $Z_{\mathcal{C}} := \sum_{i \in \mathcal{C}} \tau_i$, and $\mathbf{K} := \mathbf{H}^{(v)} (\mathbf{H}^{(v)})^{\top}$. The MMD loss expands as:

$$\mathcal{L}_{causal}^{(v)} = \frac{\mathbf{w}_{\mathcal{T}}^{\top} \mathbf{K} \mathbf{w}_{\mathcal{T}}}{Z_{\mathcal{T}}^2} - 2 \frac{\mathbf{w}_{\mathcal{T}}^{\top} \mathbf{K} \mathbf{w}_{\mathcal{C}}}{Z_{\mathcal{T}} Z_{\mathcal{C}}} + \frac{\mathbf{w}_{\mathcal{C}}^{\top} \mathbf{K} \mathbf{w}_{\mathcal{C}}}{Z_{\mathcal{C}}^2}. \quad (53)$$

For $i \in \mathcal{T}$, apply quotient rule to each term:

$$\frac{\partial}{\partial \tau_i} \left(\frac{\mathbf{w}_{\mathcal{T}}^{\top} \mathbf{K} \mathbf{w}_{\mathcal{T}}}{Z_{\mathcal{T}}^2} \right) = \frac{2[\mathbf{K} \mathbf{w}_{\mathcal{T}}]_i \cdot Z_{\mathcal{T}}^2 - \mathbf{w}_{\mathcal{T}}^{\top} \mathbf{K} \mathbf{w}_{\mathcal{T}} \cdot 2Z_{\mathcal{T}}}{Z_{\mathcal{T}}^4} \quad (54)$$

$$= \frac{2}{Z_{\mathcal{T}}^2} \left([\mathbf{K} \mathbf{w}_{\mathcal{T}}]_i - \frac{\mathbf{w}_{\mathcal{T}}^{\top} \mathbf{K} \mathbf{w}_{\mathcal{T}}}{Z_{\mathcal{T}}} \right) \quad (55)$$

$$= \frac{2}{Z_{\mathcal{T}}} (\langle \mathbf{h}_i, \boldsymbol{\mu}_{\mathcal{T}} \rangle - \|\boldsymbol{\mu}_{\mathcal{T}}\|^2), \quad (56)$$

$$-2 \frac{\partial}{\partial \tau_i} \left(\frac{\mathbf{w}_{\mathcal{T}}^{\top} \mathbf{K} \mathbf{w}_{\mathcal{C}}}{Z_{\mathcal{T}} Z_{\mathcal{C}}} \right) = -\frac{2}{Z_{\mathcal{T}} Z_{\mathcal{C}}} \left([\mathbf{K} \mathbf{w}_{\mathcal{C}}]_i - \frac{\mathbf{w}_{\mathcal{T}}^{\top} \mathbf{K} \mathbf{w}_{\mathcal{C}}}{Z_{\mathcal{T}}} \right) \quad (57)$$

$$= -\frac{2}{Z_{\mathcal{T}}} (\langle \mathbf{h}_i, \boldsymbol{\mu}_{\mathcal{C}} \rangle - \langle \boldsymbol{\mu}_{\mathcal{T}}, \boldsymbol{\mu}_{\mathcal{C}} \rangle). \quad (58)$$

Summing Eqs. (54) and (57):

$$\frac{\partial \mathcal{L}_{causal}^{(v)}}{\partial \tau_i} = \frac{2}{Z_{\mathcal{T}}} (\boldsymbol{\mu}_{\mathcal{T}} - \boldsymbol{\mu}_{\mathcal{C}})^{\top} (\mathbf{h}_i - \boldsymbol{\mu}_{\mathcal{T}}), \quad \forall i \in \mathcal{T}. \quad (59)$$

By symmetry, for $i \in \mathcal{C}$:

$$\frac{\partial \mathcal{L}_{causal}^{(v)}}{\partial \tau_i} = -\frac{2}{Z_{\mathcal{C}}} (\boldsymbol{\mu}_{\mathcal{T}} - \boldsymbol{\mu}_{\mathcal{C}})^{\top} (\mathbf{h}_i - \boldsymbol{\mu}_{\mathcal{C}}). \quad (60)$$

The total gradient is $\mathbf{g} = \sum_{v=1}^V \nabla_{\tau} \mathcal{L}_{causal}^{(v)}$.

D Related Work

In this appendix, we provide a comprehensive review of the existing literature on federated multi-view clustering (FMVC). Based on the distribution of data across clients, existing methods can be broadly categorized into horizontal FMVC, vertical FMVC, and heterogeneous/incomplete FMVC.

D.1 Horizontal Federated Multi-View Clustering

Horizontal FMVC addresses the scenario where different clients hold different sets of samples but share the same feature space (views). The primary challenge here is to aggregate cluster structures from local clients without transmitting raw sample data.

- **Consensus Matrix Factorization:** Barcena et al. proposed **FedCM/FedFCM** [1], which introduces federated versions of C-means and Fuzzy C-means algorithms for both horizontal and vertical partitioning scenarios. The key contribution is developing “lossless” federated procedures that identify exactly the same clusters as the original centralized versions while preserving data privacy. In the horizontal setting, clients share and aggregate gradient information during the iterative clustering process.
- **Prototype-based Methods:** To enhance communication efficiency in horizontal settings, Hu et al. developed **FedMVFP** [11]. This method utilizes fuzzy prototype clustering, where clients upload local prototypes (cluster centers) rather than gradients. The framework introduces a consensus prototype learning strategy that integrates clustering prototypes from different views into a unified optimization formulation, significantly reducing communication overhead.
- **Multi-View K-Means Adaptation:** Yang et al. introduced **Fed-MVKM** [19], a federated adaptation of the classic Multi-View K-Means. The method addresses the challenge of performing multi-view K-means clustering when multi-view data is distributed across multiple clients. It employs a secure aggregation mechanism to update cluster centroids iteratively while ensuring privacy preservation.

D.2 Vertical Federated Multi-View Clustering

Vertical FMVC is the most prevalent setting in federated multi-view learning, where each client (view) possesses different features for the same set of sample IDs. The core difficulty lies in aligning entities and learning cross-view correlations without exposing feature details.

- **Communication Efficiency:** A major bottleneck in vertical federated learning is communication bandwidth. Liu et al. proposed **CeFMC** [13], which addresses this by approximating data representations with pseudo-labels and centroid matrices instead of transmitting full embedding matrices. The method incorporates linear kernel functions to consider pairwise similarities and achieves linear computational complexity. Similarly, while not explicitly detailed in our references, methods like **FedMVL** [12] focus on learning common representations by exchanging high-level information rather than raw features.
- **Integrated Matrix Factorization and Clustering:** Feng et al. developed **FMVC-IMK** [8], which integrates matrix factorization and multi-view K-means into a unified optimization framework. This one-step approach avoids the performance degradation from separating feature extraction and clustering. The method employs adaptive weights to balance the influence of different views and incorporates graph-based regularization to preserve geometric structure.
- **Fuzzy Clustering with Spectral Methods: FMVFCMSP** [7] extends fuzzy C-means clustering to the federated setting by utilizing tensor Schatten- p norm regularization. The method replaces hard clustering assignments with fuzzy membership degrees, enabling better uncertainty handling and less information loss. The tensor-based regularizer fully explores inter-view complementary information and global spatial structure.
- **Scalable Spectral Clustering:** To improve scalability, Feng et al. proposed **SFOMVC-TR** [6], which achieves one-step clustering at each client and employs tensor learning to capture consistent and complementary information. The method adopts anchor graphs to enhance efficiency in high-dimensional data and incorporates $L_{p,q}$ sparse regularization on projection matrices to directly project anchors into clustering assignments.
- **Tensor-based Approaches:** Capturing high-order correlations between views is critical. Feng et al. introduced **TensorFMVC** [9], which is built on K-means and utilizes tensor factorization to model complex interactions between different views. A three-order tensor stacked by cluster assignment matrices is introduced to exploit complementary information and spatial structure. The method avoids initializing centroids to address K-means' sensitivity to initialization.
- **Information-Theoretic Feature Splitting:** Chen et al. proposed **ESFMC** [3], which addresses the inherent tension between privacy protection and clustering performance. Unlike methods that share model parameters or intermediate outputs, ESFMC adopts an information-theoretic perspective to split locally extracted features, retaining sensitive information on clients and only sharing features highly relevant to the clustering task. This privacy-preserving information sharing strategy reduces privacy risks while enabling the server to mine high-quality global clustering structures.

D.3 Heterogeneous and Incomplete FMVC

Real-world federated systems often face data challenges such as missing views (incompleteness) or inconsistent client architectures (heterogeneity).

- **Incomplete Multi-View Data:** Chen et al. addressed the incomplete data problem in **FedDMVC** [5], where views on different clients often have feature heterogeneity and data incompleteness. They proposed sample alignment and data extension techniques at the server to explore complementary cluster structures. The server distributes global prototypes and pseudo-labels as global self-supervised information, which clients use with deep autoencoders to learn view-specific cluster assignments.
- **Graph Neural Networks for Incomplete Views:** Yan et al. proposed **FIM-GNNs** [18], which addresses the challenge where different views are stored across multiple clients. The method aggregates heterogeneous features from overlapping samples into a global feature representation and generates global pseudo-labels at the server to enhance handling of incomplete view data.
- **Heterogeneous Hybrid Views:** Chen et al. tackled the challenge of heterogeneous hybrid views in **FMCS** [4], where a mixture of single-view and multi-view clients exhibit varying degrees of heterogeneity. They introduced a local-synergistic contrastive learning approach to achieve consistency among heterogeneous clients and developed a global-specific weighting aggregation method to learn complementary features from hybrid views.
- **Unaligned and Incomplete Data:** Ren et al. proposed **FCUIF** [15], which eliminates the strict assumptions of data alignment and completeness prevalent in existing FedMVC methods. To address unaligned multi-view data, FCUIF leverages both sample commonality and view versatility to adaptively generate alignment matrices for effective cross-view alignment. For incomplete data, it employs an unsupervised technique to evaluate and refine imputation quality, efficiently handling various scenarios of missing multi-view data.

D.4 Summary and Discussion

While the aforementioned methods have successfully addressed privacy, efficiency, and data imperfection challenges, they share a common limitation: **they fundamentally rely on statistical correlations**. Whether employing matrix factorization, tensor decomposition, fuzzy clustering, or deep learning, these models aim to maximize the association between features and cluster assignments.

However, as highlighted in causal learning theory, statistical correlation does not imply causation. In the distributed setting of FMVC, data is often collected under different selection biases (confounders), leading to *spurious correlations*. For instance, a background feature might be strongly correlated with a label in one client's dataset due to sampling bias, but this correlation is non-causal. Existing FMVC methods lack the mechanism to distinguish between causal features and confounder-driven noise. This observation motivates our proposed method, which integrates causal inference—specifically confounder balancing—into the FMVC framework to achieve robust and “spurious-free” clustering.

E Algorithm Details

This section presents the complete pseudocode for SFFedMC, integrating the Block Coordinate Ascent with Causal Re-weighting (BCA-CR) optimization developed in Section 3.3 of the main text. Algorithm 1 describes the initialization, client-side representation updates via Orthogonal Procrustes, server-side consensus aggregation, and the core causal weight balancing mechanism.

E.1 Detailed Description of Execution Steps

The execution workflow of SFFedMC, as outlined in Algorithm 1, proceeds in two main stages: a lightweight initialization followed by an alternating optimization loop.

Initialization (Lines 1–10). To facilitate fast convergence, SFFedMC employs a warm-start strategy. First, the global causal weights τ are initialized uniformly as $1/N$, assuming no prior knowledge of confounder bias (Line 1). Simultaneously, each client performs a local Singular Value Decomposition (SVD) on its raw features $\mathbf{X}^{(v)}$ to extract initial low-rank representations $\mathbf{H}^{(v)}$ (Lines 4–5). These initial embeddings are masked (to protect privacy) and sent to the server to generate coarse-grained global centroids $\mathbf{C}^{(v)}$, and consensus labels $\mathbf{y}_c$ via a one-shot K-Means clustering (Line 9).

Phase 1: Client-Side Local Update (Lines 13–26). In each iteration, clients perform computations in parallel to refine their local views.

- **Representation Learning:** The core step is updating the latent representation $\mathbf{H}^{(v)}$. As shown in Lines 17–21, this is formulated as an Orthogonal Procrustes problem. We construct an auxiliary matrix $\mathbf{M}$ that combines the *causally weighted* kernel information ($\Gamma(\tau)\mathbf{K}^{(v)}\mathbf{H}^{(v)}$) with the structural guidance from local clusters ($\lambda\mathbf{Y}^{(v)}\mathbf{C}^{(v)}$). The optimal $\mathbf{H}^{(v)}$ is then obtained via SVD on $\mathbf{M}$, ensuring orthogonality constraints are strictly met.
- **Label Assignment:** Clients update their local pseudo-labels $\mathbf{y}^{(v)}$ by maximizing the alignment score S , which balances the agreement between the local representation and both local and global centroids (Lines 23–24).

Phase 2: Server-Side Global Aggregation (Lines 27–43). The server coordinates the global consensus and, crucially, performs the causal intervention.

- **Consensus Formation:** The server aggregates the uploaded local information to form a consensus matrix $\mathbf{H}_c$. It then updates the global centroids $\mathbf{C}_c$ via SVD (Lines 29–31) and determines the global cluster assignments $\mathbf{y}_c$ using a weighted voting mechanism, where samples with higher causal weights τ have more influence (Lines 33–34).
- **Causal Weight Update (Key Innovation):** Lines 36–40 implement the proposed confounder balancing. The server calculates the gradient of the Multi-View Maximum Mean Discrepancy (MMD) loss. Specifically, for each view, it partitions samples into treated and control groups based on the current representation. It then updates τ to minimize the distributional distance between these groups.
- **Projection:** Finally, the updated weights are projected onto the probability simplex (Line 42) to ensure they remain non-negative and sum to one, maintaining their probabilistic interpretation.

This alternating procedure continues until the change in the objective function falls below a threshold or the maximum iteration count is reached, guaranteeing a stable and spurious-free clustering result.

F Additional Experimental Results

F.1 Detailed Description of Datasets

Table 2 provides detailed statistics of the eight multi-view datasets used in our experiments.

Table 2: Statistics of multi-view datasets. Each dataset is characterized by domain, number of views (#V), number of samples (#N), number of classes (#K), and feature dimensions per view.

Dataset	Domain	#V	#N	#K	Dimensions
Caltech101-7	Image	6	1474	7	[48, 40, 254, 1984, 512, 928]
Hdigit	Image	2	10000	10	[784, 256]
Mfeat	Image	6	2000	10	[216, 76, 64, 6, 240, 47]
Multi-Fashion	Image	3	10000	10	[784, 784, 784]
MSRC-v1	Image	4	210	7	[24, 512, 256, 254]
ORL	Image	3	400	40	[4096, 3304, 6750]
WebKB	Text	3	203	4	[1703, 230, 230]
Yale	Image	3	165	15	[4096, 3304, 6750]

Caltech101-7¹ is an image dataset containing 7 categories (faces, motorbikes, dollar bills, garfields, snoopy, windsor chairs, leopards) with 6 visual feature views extracted using different descriptors.

¹<https://www.kaggle.com/datasets/imbikramsaha/caltech-101>.

Hdigit² contains handwritten digit images (0-9) represented by 2 views: pixel features and Fourier coefficients.

Mfeat³ provides handwritten digits with 6 feature types: Fourier coefficients, profile correlations, Karhunen-Love coefficients, pixel averages, Zernike moments, and morphological features.

Multi-Fashion⁴ extends Fashion-MNIST with 3 views capturing different visual perspectives of fashion items.

MSRC-v1⁵ is an image segmentation dataset with 7 object categories represented by 4 visual feature views.

ORL⁶ contains face images of 40 individuals with 3 feature views: intensity, LBP, and Gabor descriptors.

WebKB⁷ is a webpage classification dataset with 4 categories represented by 3 textual views: page content, inlinks, and outlinks.

Yale⁸ contains face images of 15 individuals under various lighting conditions, represented by 3 feature views similar to ORL.

F.2 Implementation Details

All experiments were conducted on a workstation equipped with an Intel Core Ultra 7 265K processor, NVIDIA RTX 5070 Ti GPU (16GB VRAM), and 48GB system memory. We implemented SFFedMC in Python 3.12 using NumPy 2.3.5 and SciPy 1.16.3 for numerical computations. For baseline methods, we used the official implementations (except the FMVFCMSP method) provided by the authors with default hyperparameters recommended in their papers.

- **CeFMC**: https://github.com/liujiyuan13/CeFMC-code_release.
- **FedCMv**: <https://github.com/Unipisa/FederatedClustering>.
- **FedFCMv**: <https://github.com/Unipisa/FederatedClustering>.
- **FedMVL**: <https://github.com/Mycroft-s/Vertical-Federated-MVML-methods/tree/main/FedMVL>.
- **ESFMC**: <https://github.com/5Martina5/ESFMC>.
- **FCUIF**: <https://github.com/5Martina5/FCUIF>.
- **FMVFCMSP**: We reimplemented this method following the algorithmic details described in the original paper [7], as the authors have not released the source code.

Data Preprocessing. Following standard practice in multi-view clustering, we apply L2 normalization to each sample across all views to ensure features are on comparable scales. For datasets with high-dimensional features (e.g., ORL, Yale), we compute the kernel matrix $\mathbf{K}^{(v)} = \mathbf{X}^{(v)}(\mathbf{X}^{(v)})^\top$ once and reuse it throughout iterations to reduce computational overhead.

Hyperparameter Selection. We set the subspace dimension $c = K$ (number of clusters) for all datasets, following prior work [13]. The regularization parameters are fixed as $\lambda = 1.0$, $\beta = 1.0$, $\gamma = 0.01$ across all experiments to demonstrate robustness. The learning rate for causal weight update is $\eta = 0.01$. Maximum outer iterations $T_{\max} = 100$ with convergence tolerance $\epsilon = 10^{-5}$; inner iterations for client-side updates $T_{\text{inner}} = 10$.

F.3 Additional Results on Ablation Study

Figures 1, 2, and 3 present ablation results for NMI, F-score, and ARI, respectively. Consistent with ACC findings, the full SFFedMC consistently outperforms all ablated variants across all metrics and datasets. The degradation occurs when removing causal regularization (NoCR), particularly on datasets with strong confounder effects (e.g., Yale, WebKB). This confirms that the proposed causal weight balancing mechanism is critical for robust clustering in federated settings.

F.4 Additional Results on Convergence Analysis

Figures 4 through 10 show convergence behavior across all datasets. In every case, the objective function stabilizes within 2-4 iterations, and the relative change drops below the tolerance threshold rapidly. The shaded regions represent standard deviations across 5 runs, indicating consistent convergence behavior. This demonstrates that the BCA-CR optimization is efficient and robust across diverse data distributions.

F.5 Additional Results on Parameter Sensitivity

Figures 11, 12, and 13 present parameter sensitivity analyses for NMI, F-score, and ARI, respectively. The results confirm that SFFedMC maintains stable performance across a wide range of hyperparameter values for all metrics. Similar to ACC findings, only WebKB exhibits slight sensitivity to extreme parameter settings ($\lambda < 0.1$, $\beta > 30$, $\gamma > 0.3$). For practical applications, we recommend setting $\lambda \in [0.1, 10]$, $\beta \in [0.1, 10]$, and $\gamma \in [0.001, 0.1]$.

²https://github.com/ManshengChen/Multi-view_Datasets.

³<https://archive.ics.uci.edu/dataset/72/multiple+features>.

⁴<https://github.com/zalandoresearch/fashion-mnist>.

⁵https://github.com/ManshengChen/Multi-view_Datasets.

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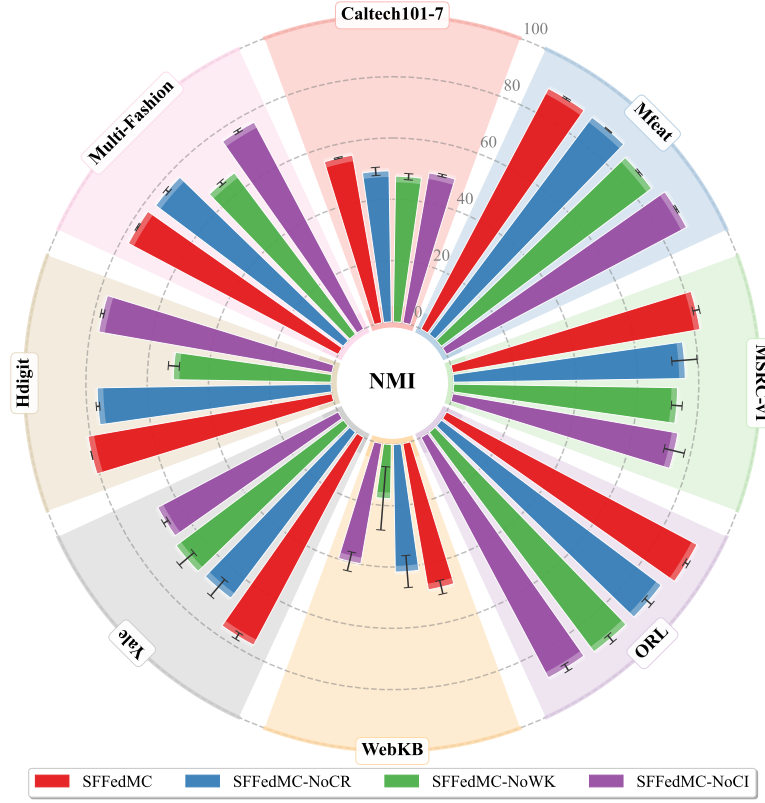


Figure 1: Ablation study on NMI. The full SFFedMC achieves the best performance across all datasets.

F.6 Additional Results on Visualization Analysis

Figures 14 through 20 present t-SNE visualizations and weight distributions for all datasets. In every case, the learned consensus representations exhibit clear cluster separation in the embedding space, validating the effectiveness of the federated optimization. The causal weight distributions reveal that SFFedMC adaptively assigns higher weights to samples with clearer cluster membership, effectively downweighting samples affected by confounders. This adaptive weighting mechanism is crucial for achieving spurious-free clustering in heterogeneous federated environments.

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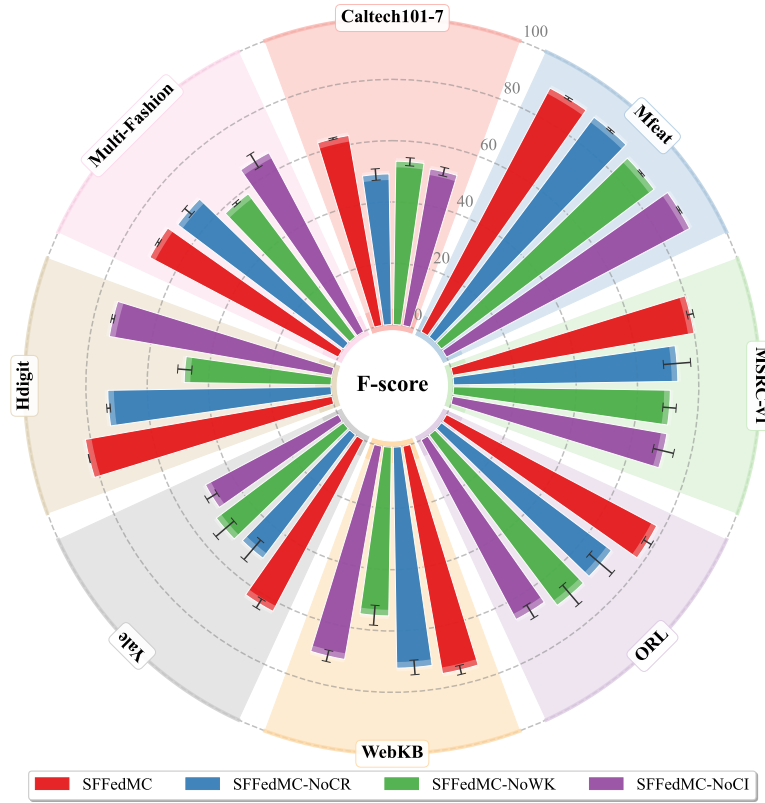


Figure 2: Ablation study on F-score. Removing causal regularization leads to the performance drop.

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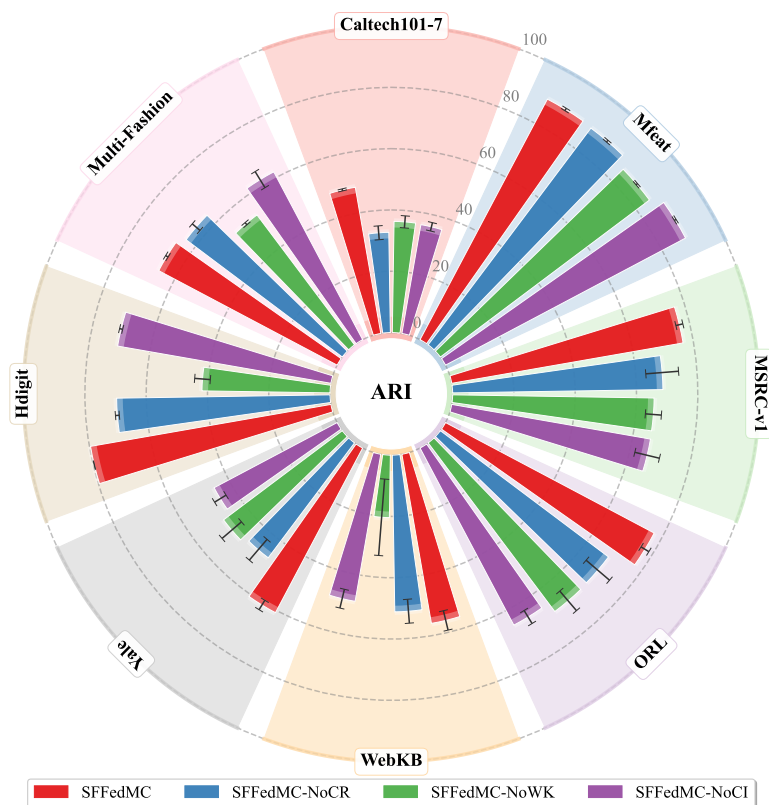


Figure 3: Ablation study on ARI. All components contribute positively to clustering quality.

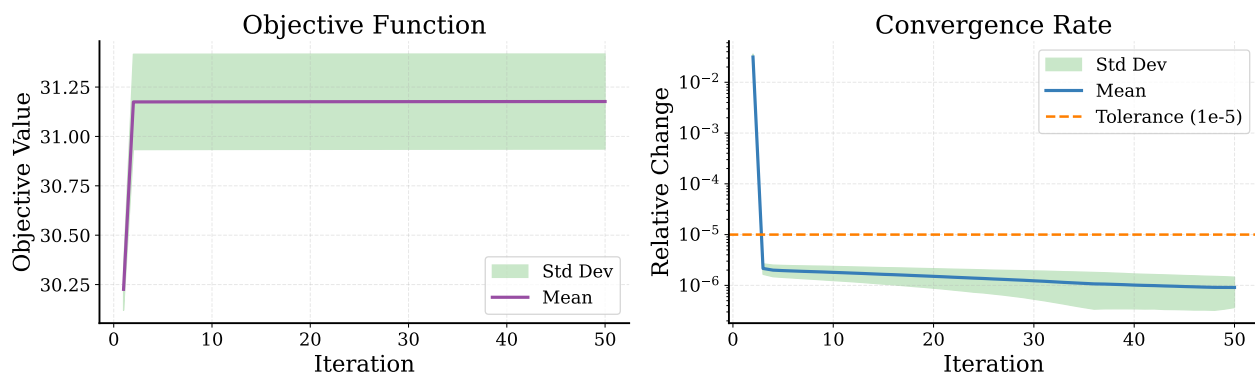


Figure 4: Convergence analysis on Caltech101-7.

Algorithm 1 SFFedMC: Spurious-Free Federated Multi-View Clustering

Input: Distributed data views $\{\mathbf{X}^{(v)}\}_{v=1}^V$; Number of clusters K ; Subspace dimension c ; Hyper-parameters λ, β, γ ; Learning rate η .

Output: Consensus cluster labels $\mathbf{y}_c$.

```

1: /* ----- Initialization Phase ----- */
2:  $\boldsymbol{\tau} \leftarrow [\frac{1}{N}, \dots, \frac{1}{N}]^\top$  ▷ Initialize causal weights uniformly
3: for  $v = 1$  to  $V$  in parallel do
4:    $\mathbf{U}, \mathbf{S}, \mathbf{V} \leftarrow \text{SVD}(\mathbf{X}^{(v)}, c)$  ▷ Local initialization
5:    $\mathbf{H}^{(v)} \leftarrow \mathbf{U}$ 
6:   Send  $\mathbf{H}^{(v)}$  (masked) to Server
7: end for
8: Server:  $\mathbf{H}_{cat} \leftarrow [\mathbf{H}^{(1)}, \dots, \mathbf{H}^{(V)}]$ 
9: Server:  $\mathbf{y}_c, \{\mathbf{C}^{(v)}\}_{v=1}^V \leftarrow \text{KMeans}(\mathbf{H}_{cat}, K)$ 
10: Server: Broadcast  $\mathbf{y}_c, \mathbf{C}^{(v)}, \boldsymbol{\tau}$  to clients

11: /* ----- Alternating Optimization ----- */
12: while not converged do
13:   // Phase 1: Client-Side Update (Parallel)
14:   for  $v = 1$  to  $V$  in parallel do
15:      $\mathbf{K}^{(v)} \leftarrow \mathbf{X}^{(v)}(\mathbf{X}^{(v)})^\top$  ▷ Compute/Load local kernel
16:      $\boldsymbol{\Gamma}(\boldsymbol{\tau}) \leftarrow \text{diag}(\boldsymbol{\tau})$ 
17:     /* Update Representation  $\mathbf{H}^{(v)}$  */
18:     for  $\text{iter} = 1$  to  $T_{inner}$  do
19:        $\mathbf{M} \leftarrow \boldsymbol{\Gamma}(\boldsymbol{\tau}) \mathbf{K}^{(v)} \mathbf{H}^{(v)} + \lambda \mathbf{Y}^{(v)} \mathbf{C}^{(v)}$ 
20:        $\mathbf{U}, \sim, \mathbf{V}^\top \leftarrow \text{SVD}(\mathbf{M}, c)$ 
21:        $\mathbf{H}^{(v)} \leftarrow \mathbf{U} \mathbf{V}^\top$  ▷ Solve orthogonal Procrustes problem
22:     end for
23:     /* Update Local Pseudo-labels  $\mathbf{y}^{(v)}$  */
24:      $\mathbf{S} \leftarrow \lambda \mathbf{H}^{(v)} (\mathbf{C}^{(v)})^\top + \beta \mathbf{C}_c [\mathbf{y}_c] (\mathbf{C}^{(v)})^\top$ 
25:      $\mathbf{y}^{(v)} \leftarrow \text{argmax}(\mathbf{S}, \text{axis} = 1)$ 
26:     Upload  $\mathbf{y}^{(v)}$  and  $\mathbf{C}^{(v)}$  to Server
27:   end for
28:   // Phase 2: Server-Side Update
29:   Server: Aggregate consensus matrix  $\mathbf{H}_c \leftarrow [\mathbf{Y}^{(1)} \mathbf{C}^{(1)}, \dots, \mathbf{Y}^{(V)} \mathbf{C}^{(V)}]$ 
30:   /* Update Global Centroids  $\mathbf{C}_c$  */
31:    $\mathbf{M}_{cc} \leftarrow (\mathbf{H}_c)^\top \mathbf{Y}_c$ 
32:    $\mathbf{U}, \sim, \mathbf{V}^\top \leftarrow \text{SVD}(\mathbf{M}_{cc}, K)$ 
33:    $\mathbf{C}_c \leftarrow \mathbf{V} \mathbf{U}^\top$ 
34:   /* Update Consensus Labels  $\mathbf{y}_c$  */
35:    $\mathbf{Scores} \leftarrow \mathbf{H}_c (\mathbf{C}_c)^\top$ 
36:    $\mathbf{y}_c \leftarrow \text{argmax}(\text{diag}(\boldsymbol{\tau}) \cdot \mathbf{Scores}, \text{axis} = 1)$  ▷ Weighted voting
37:   /* Update Sample Weights  $\boldsymbol{\tau}$  (Causal Regularization) */
38:    $\mathbf{g} \leftarrow \mathbf{0}$ 
39:   for  $v = 1$  to  $V$  do
40:     Define groups  $\mathcal{T}_v, \mathcal{C}_v$  via split on  $\mathbf{H}^{(v)}$ 
41:      $\mathbf{g} \leftarrow \mathbf{g} + \nabla_{\boldsymbol{\tau}} \|\boldsymbol{\mu}_{\mathcal{T}_v}(\boldsymbol{\tau}) - \boldsymbol{\mu}_{\mathcal{C}_v}(\boldsymbol{\tau})\|_{\mathcal{H}^{(v)}}^2$  ▷ Compute MMD gradient
42:   end for
43:    $\boldsymbol{\tau} \leftarrow \boldsymbol{\tau} - \eta \cdot \mathbf{g}$ 
44:    $\boldsymbol{\tau} \leftarrow \text{ProjectSimplex}(\boldsymbol{\tau})$  ▷ Ensure  $\sum \tau_i = 1, \tau_i \geq 0$ 
45:   Broadcast  $\mathbf{y}_c, \mathbf{C}_c, \boldsymbol{\tau}$  to clients
46: end while
47: return  $\mathbf{y}_c$ 

```

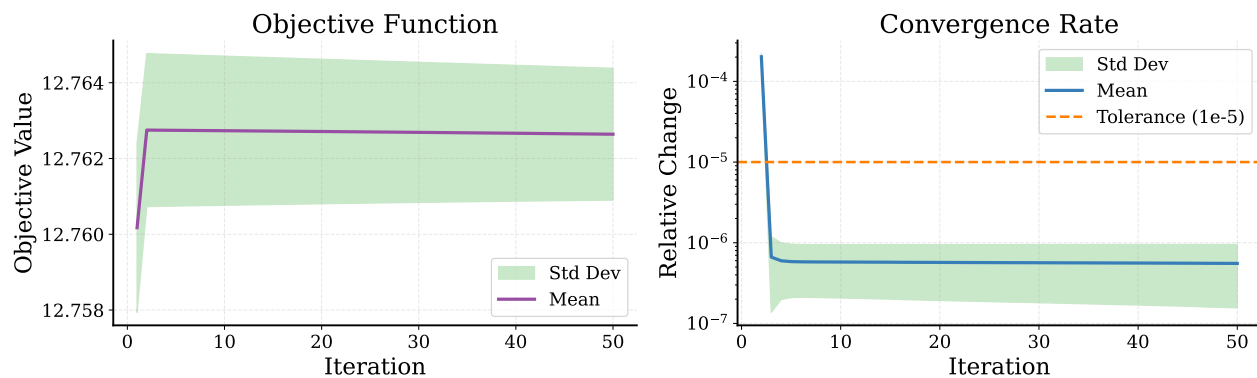


Figure 5: Convergence analysis on Hdigit.

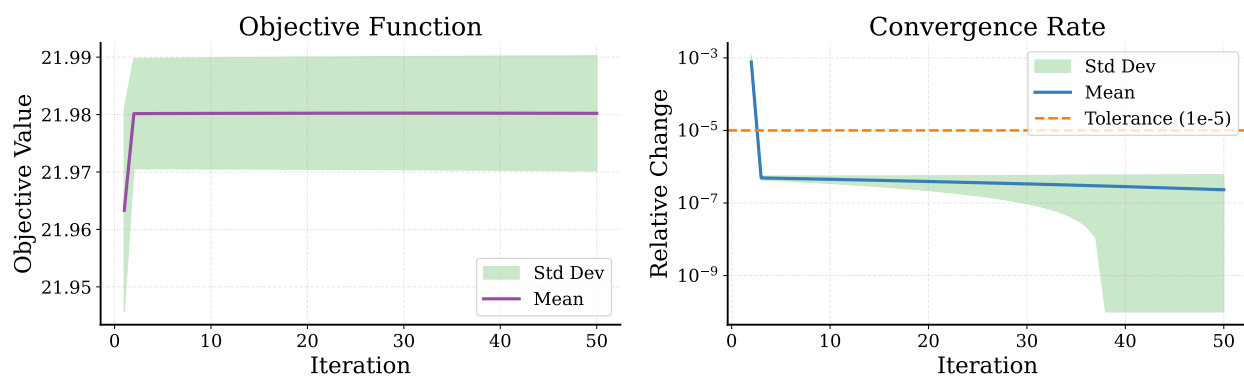


Figure 6: Convergence analysis on Multi-Fashion.

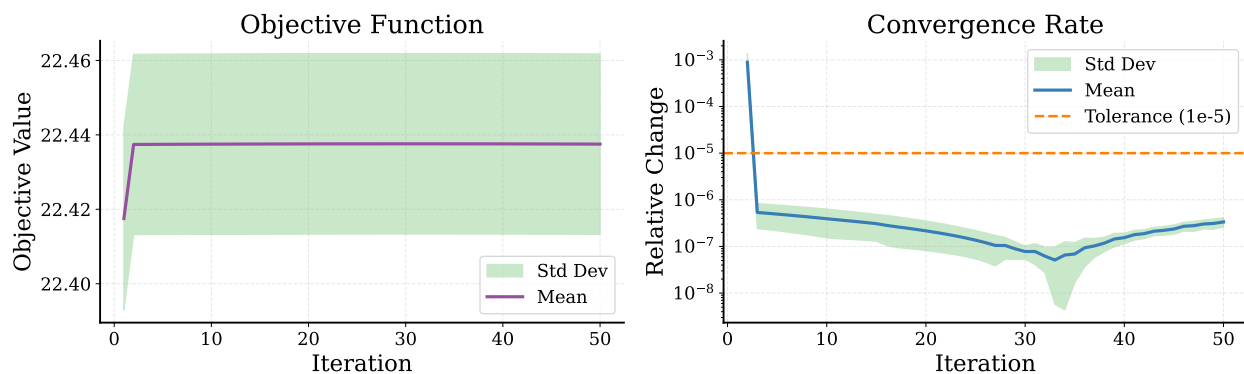


Figure 7: Convergence analysis on MSRC-v1.

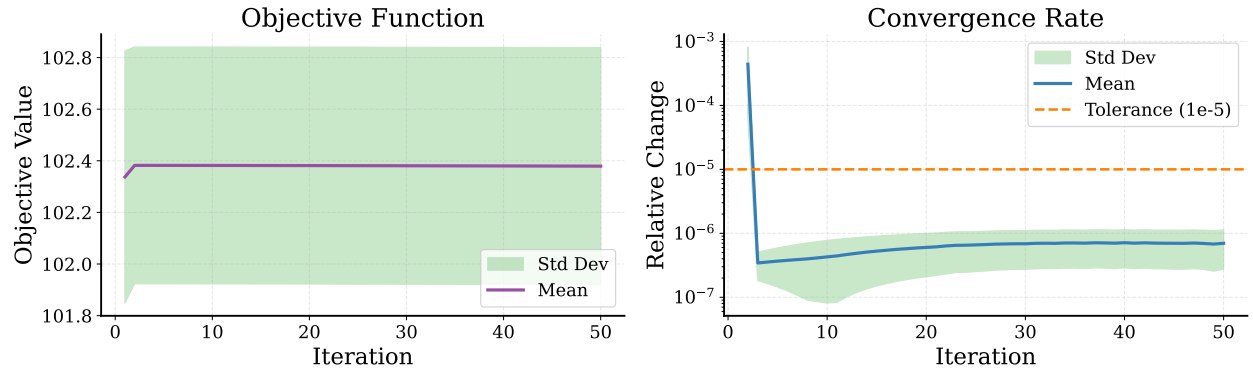


Figure 8: Convergence analysis on ORL.

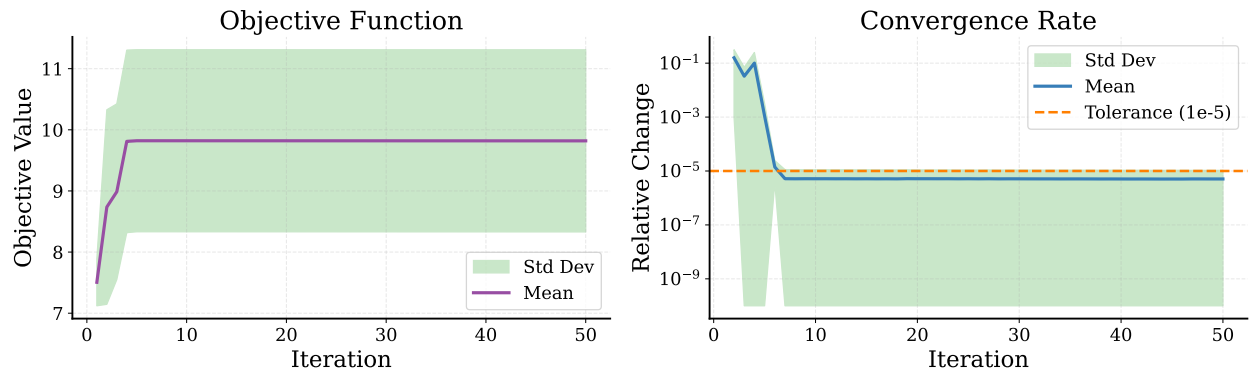


Figure 9: Convergence analysis on WebKB.

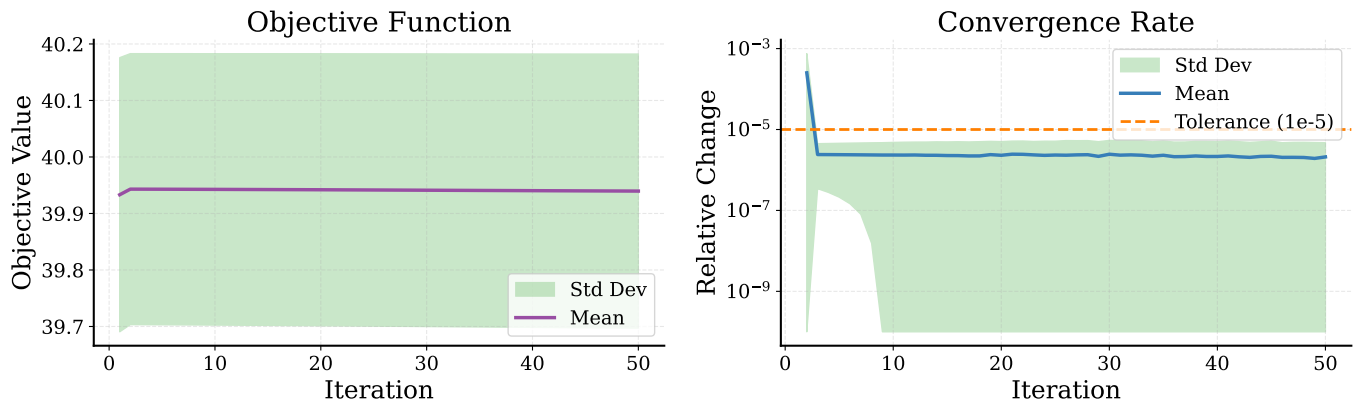


Figure 10: Convergence analysis on Yale.

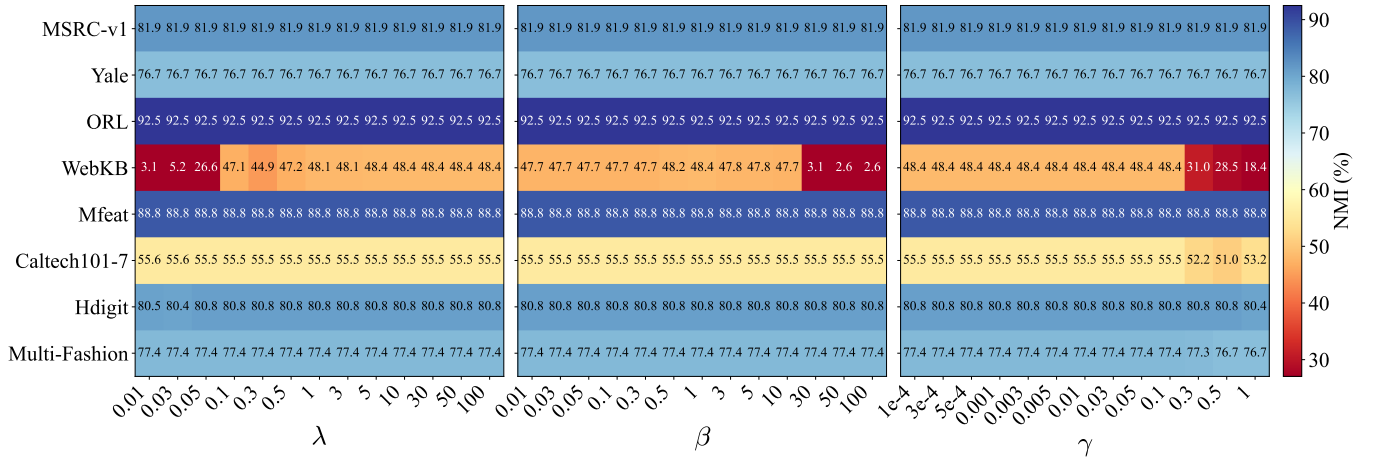


Figure 11: Parameter sensitivity analysis on NMI. SFFedMC exhibits robust performance across hyperparameter ranges.

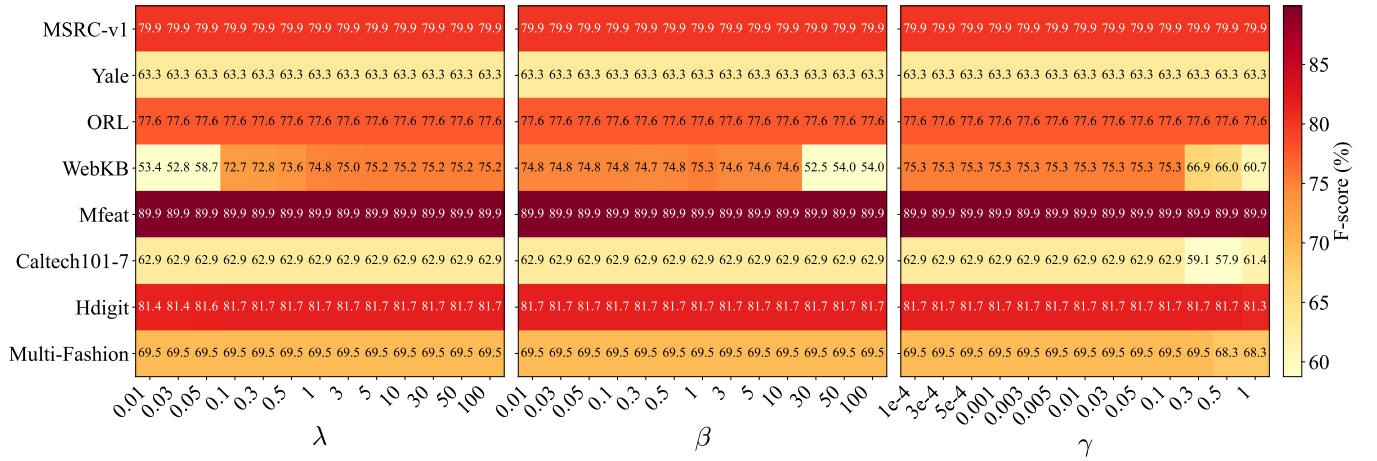


Figure 12: Parameter sensitivity analysis on F-score. Only WebKB shows slight sensitivity to extreme values.

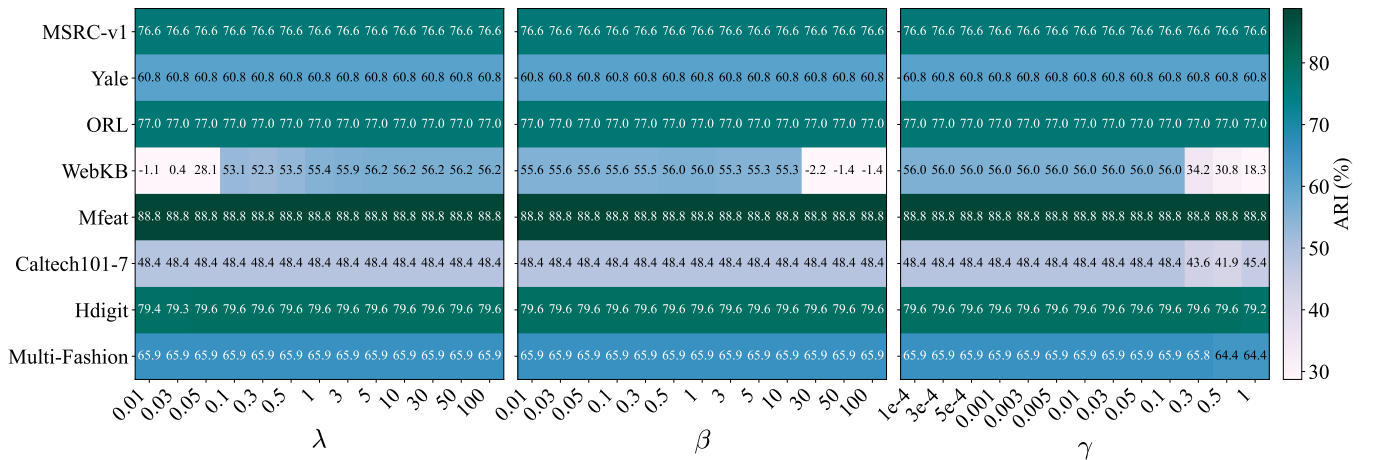
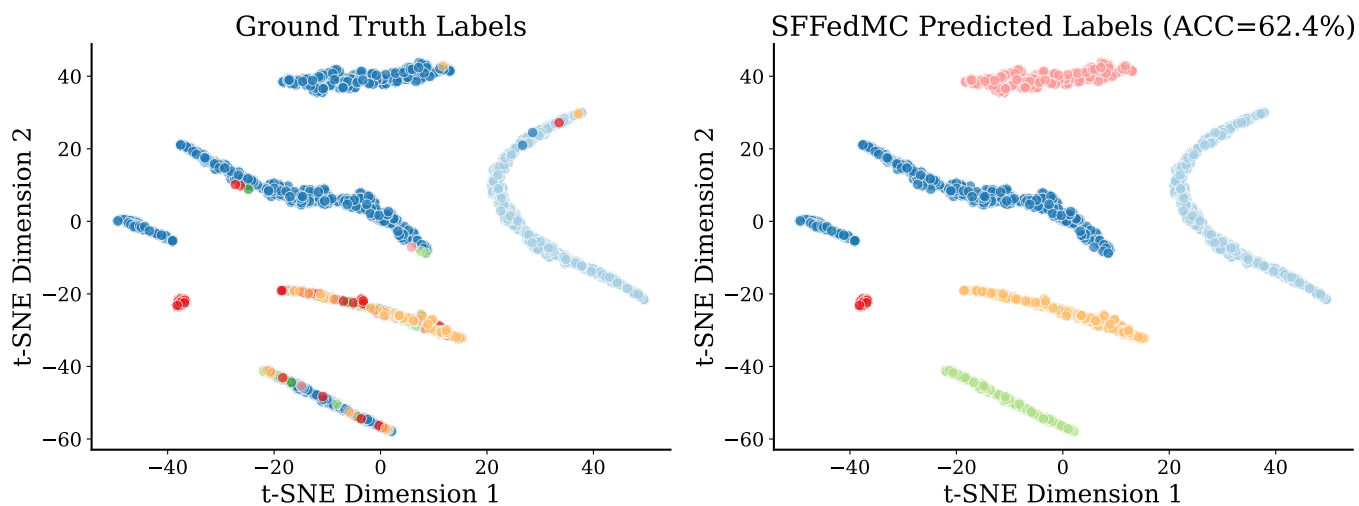


Figure 13: Parameter sensitivity analysis on ARI. Overall stable performance validates the robustness of SFFedMC.

t-SNE Visualization - Caltech101-7



Sample Weights Analysis - Caltech101-7

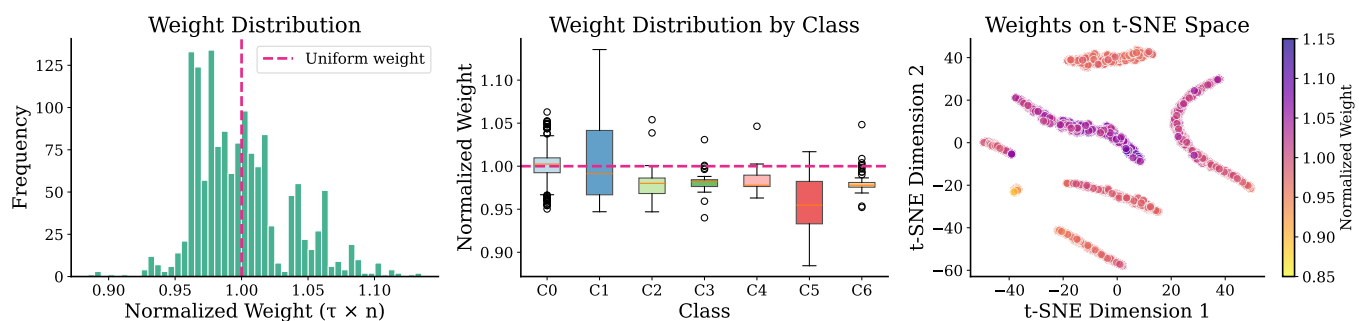
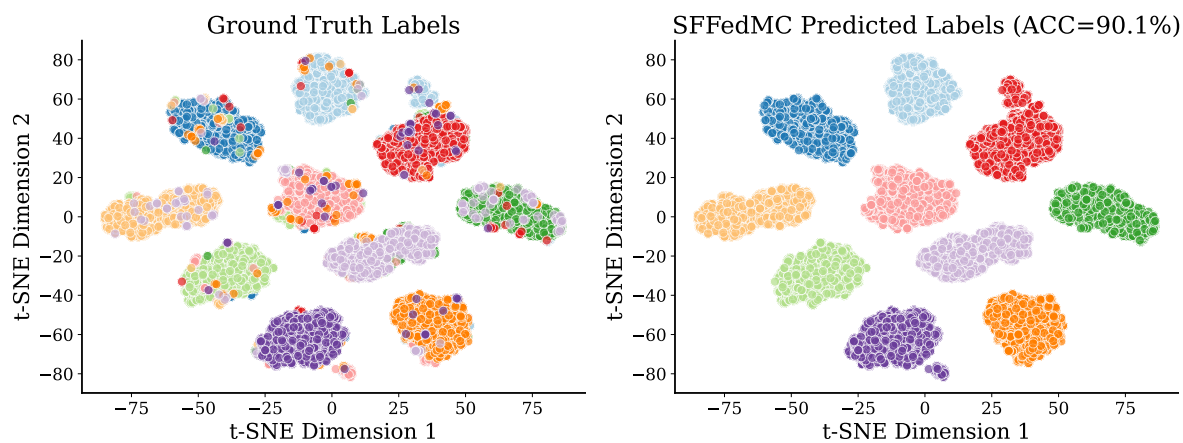


Figure 14: Visualization analysis on Caltech101-7.

t-SNE Visualization - Hdigit



Sample Weights Analysis - Hdigit

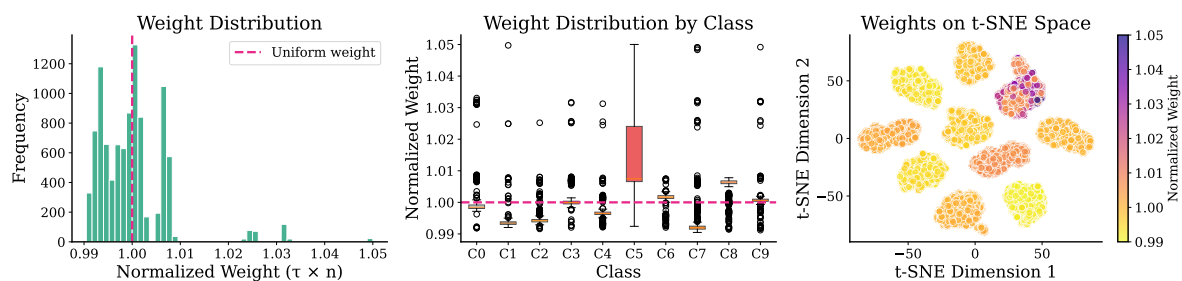
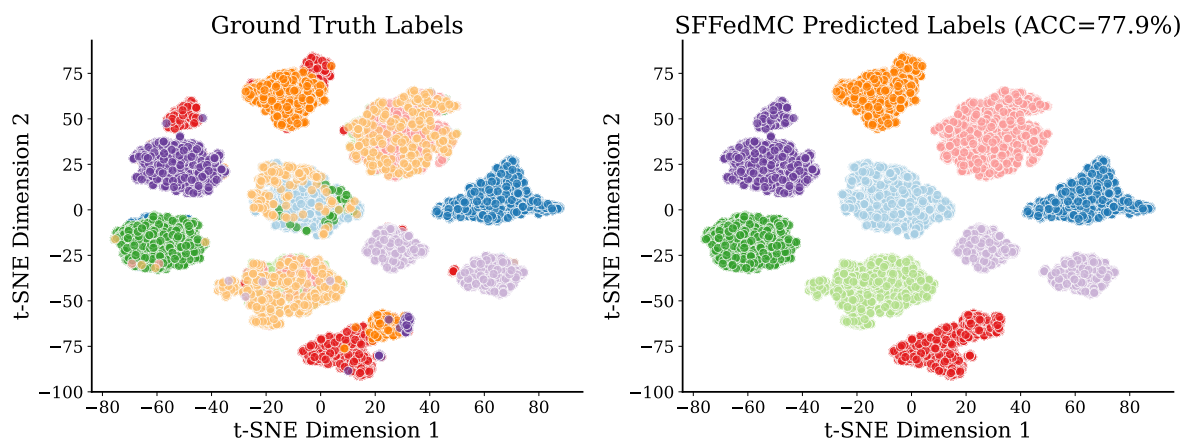
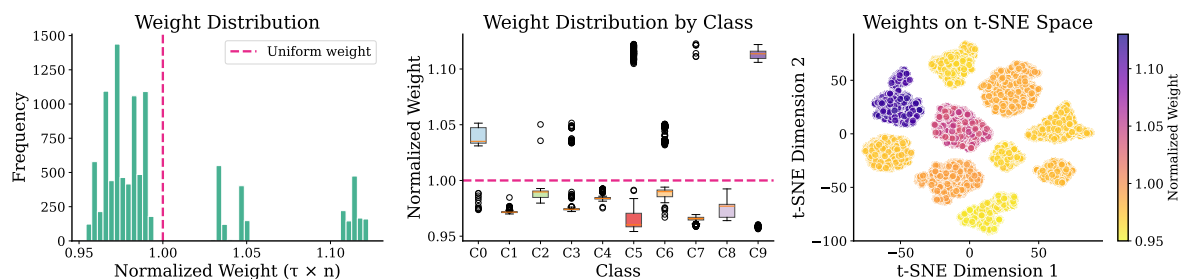
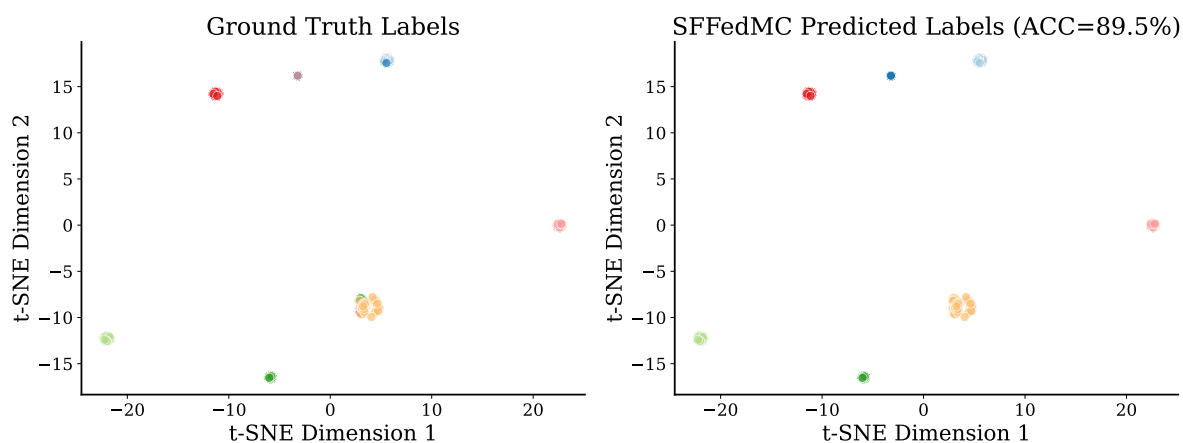


Figure 15: Visualization analysis on Hdigit.

t-SNE Visualization - Multi-Fashion**Sample Weights Analysis - Multi-Fashion****Figure 16: Visualization analysis on Multi-Fashion.**

t-SNE Visualization - MSRC-v1



Sample Weights Analysis - MSRC-v1

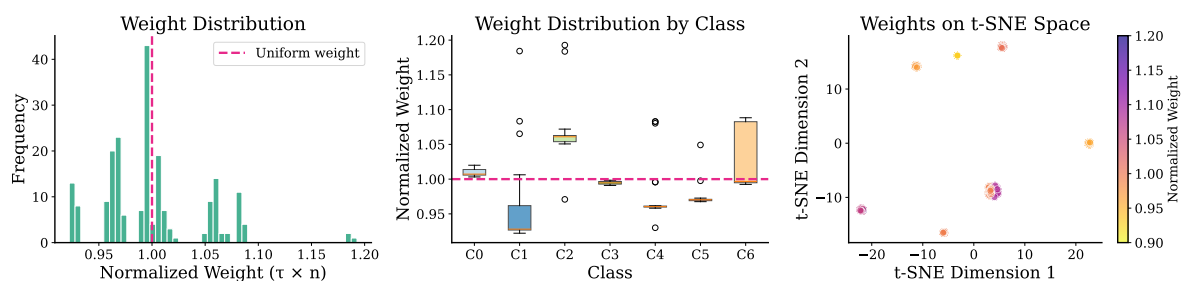
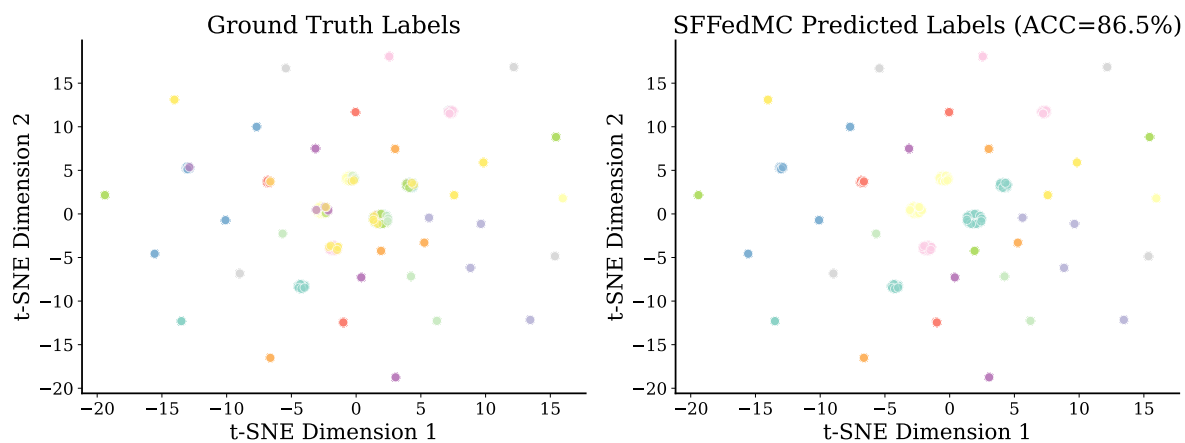
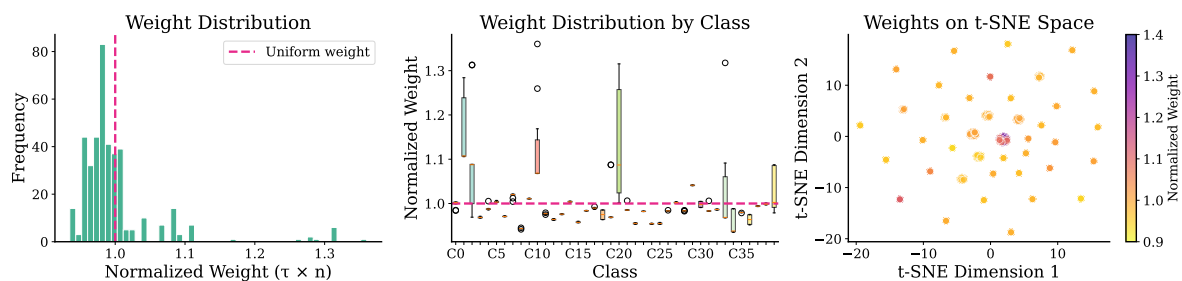
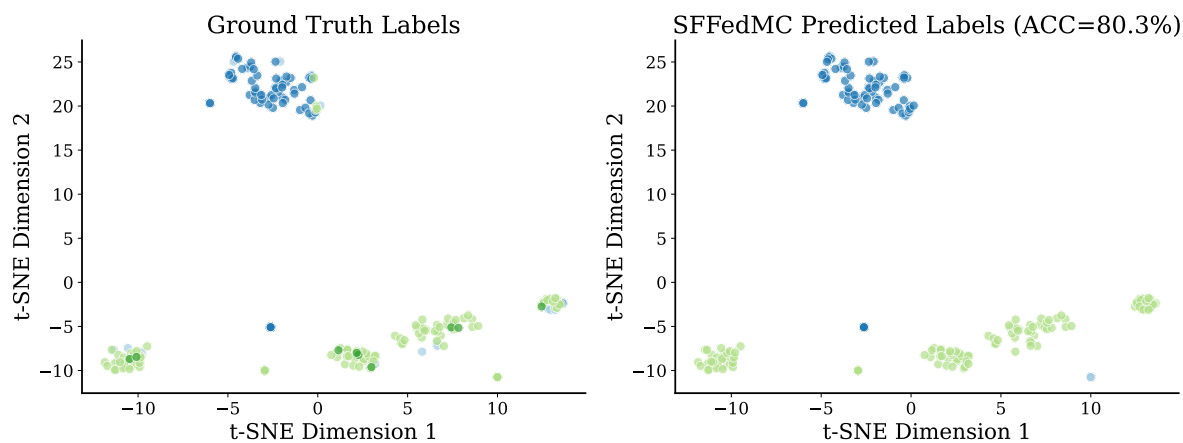


Figure 17: Visualization analysis on MSRC-v1.

t-SNE Visualization - ORL**Sample Weights Analysis - ORL****Figure 18: Visualization analysis on ORL.**

t-SNE Visualization - WebKB



Sample Weights Analysis - WebKB

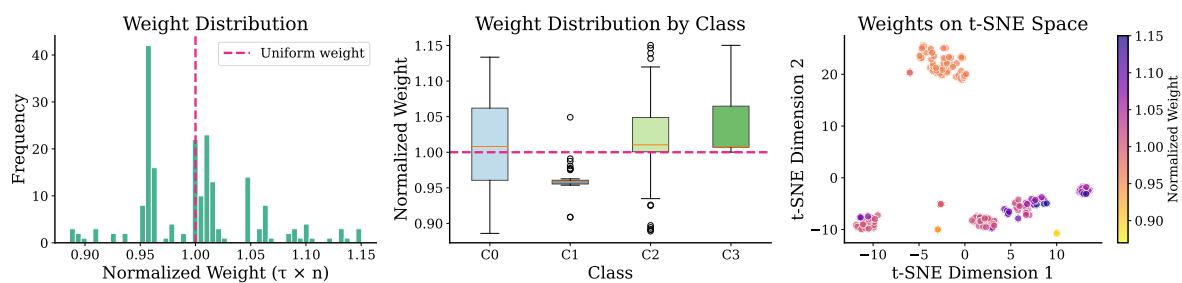
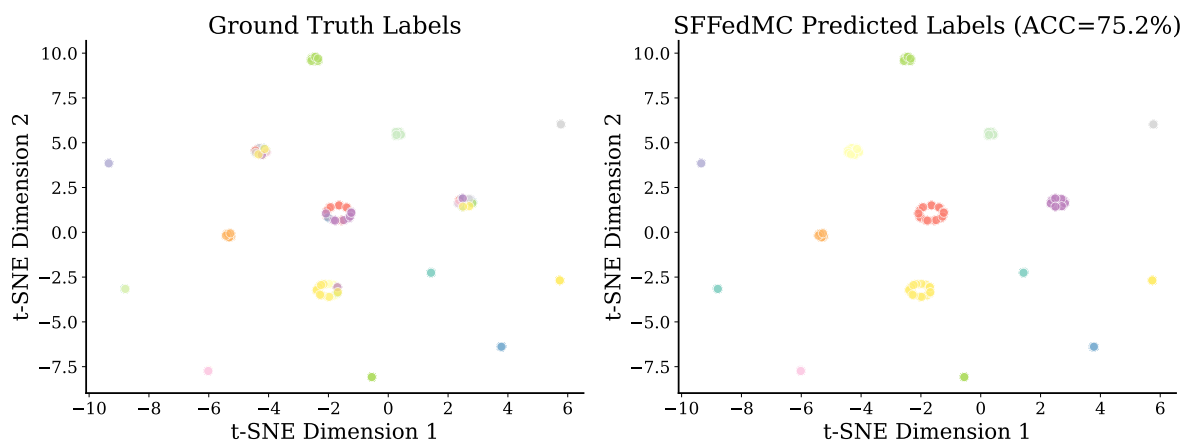
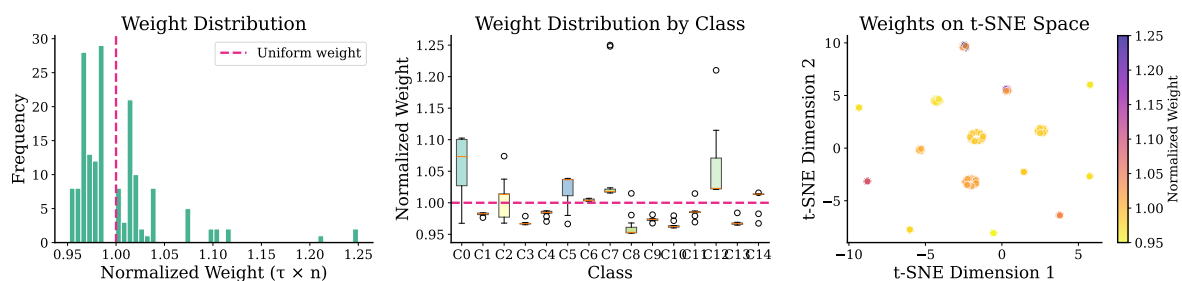


Figure 19: Visualization analysis on WebKB.

t-SNE Visualization - Yale**Sample Weights Analysis - Yale****Figure 20: Visualization analysis on Yale.**